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(54) SUPERCONDUCTING WIRE STRUCTURE, SUPERCONDUCTING COIL, AND SUPERCONDUCTING DEVICE

(71) Applicant: **KABUSHIKI KAISHA TOSHIBA**, Tokyo (JP)

(72) Inventors: **Keiko ALBESSARD**, Yokohama Kanagawa (JP); **Yasushi HATTORI**, Kawasaki Kanagawa (JP); **Tomoko EGUCHI**, Shinagawa Tokyo (JP)

(73) Assignee: **KABUSHIKI KAISHA TOSHIBA**, Tokyo (JP)

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(57) **ABSTRACT**

A superconducting wire structure of embodiments includes: a first superconducting wire; a second superconducting wire adjacent to the first superconducting wire; a joint layer electrically connecting the first superconducting wire and the second superconducting wire; and a connecting member connecting the joint layer and at least one of the first superconducting wire and the second superconducting wire. The connecting member has a melting point equal to or more than 900° C. and equal to or less than 1100° C.

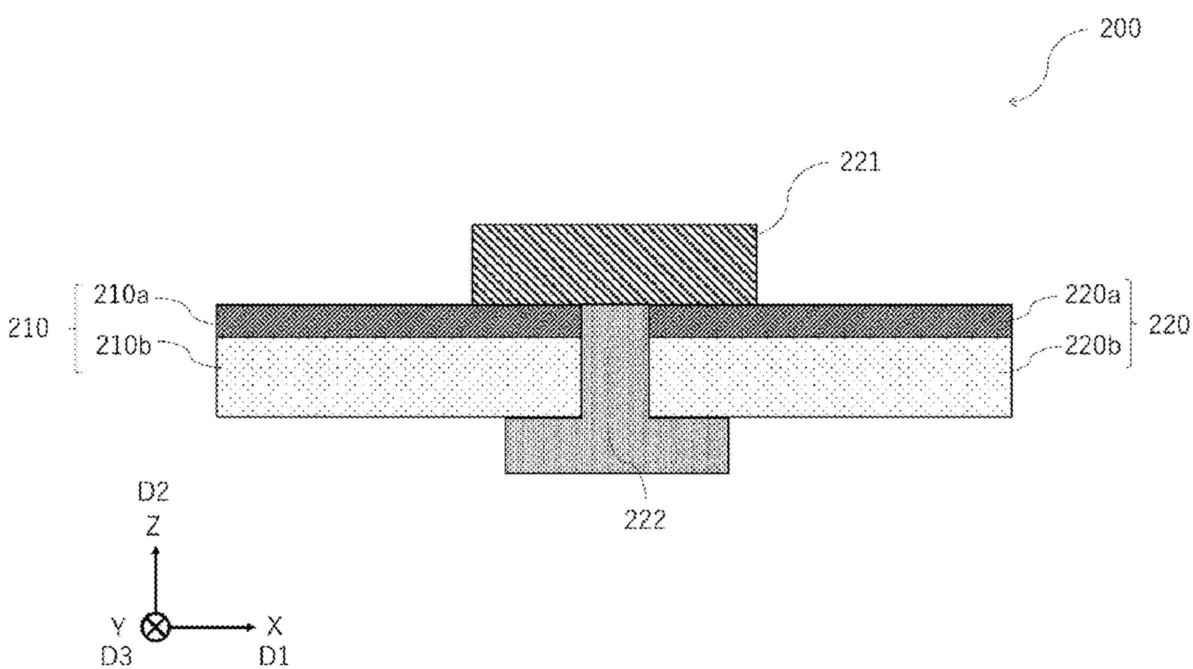


FIG. 1

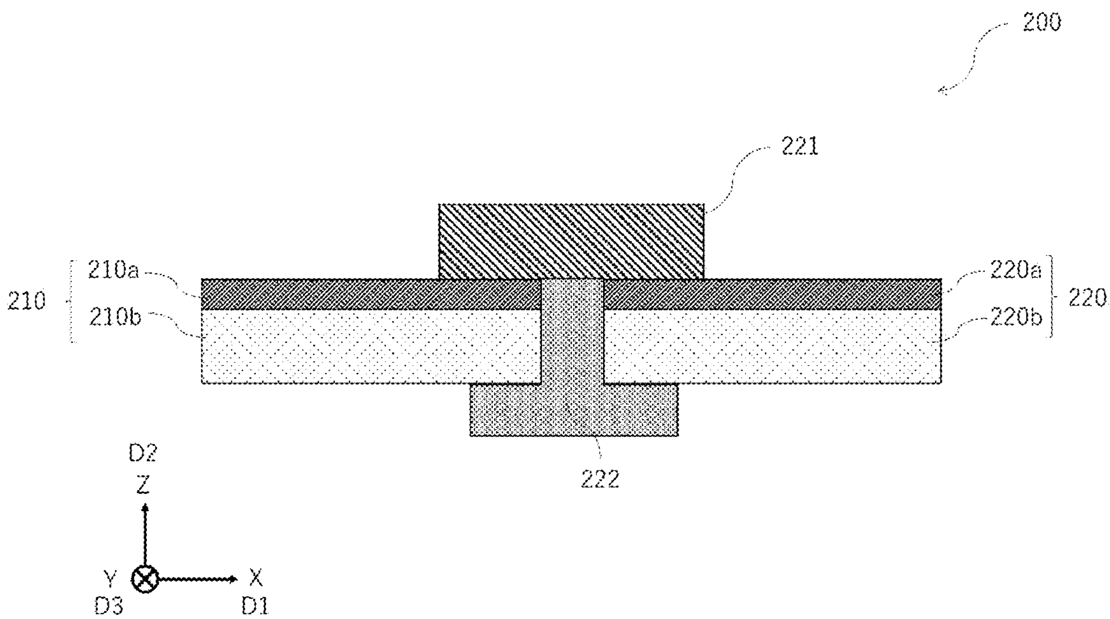

200
221
210a
210
210b
220a
220
220b
222
D2
Z
Y
X
D3
D1

FIG. 2

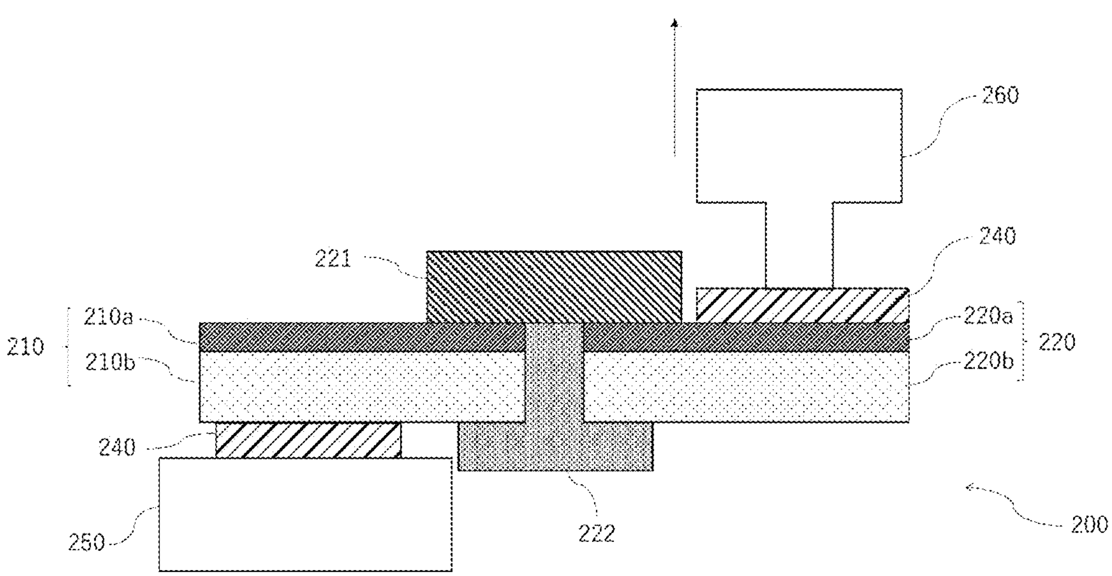

260
221
240
210a
210
210b
220a
220
220b
240
250
222
200

FIG. 3

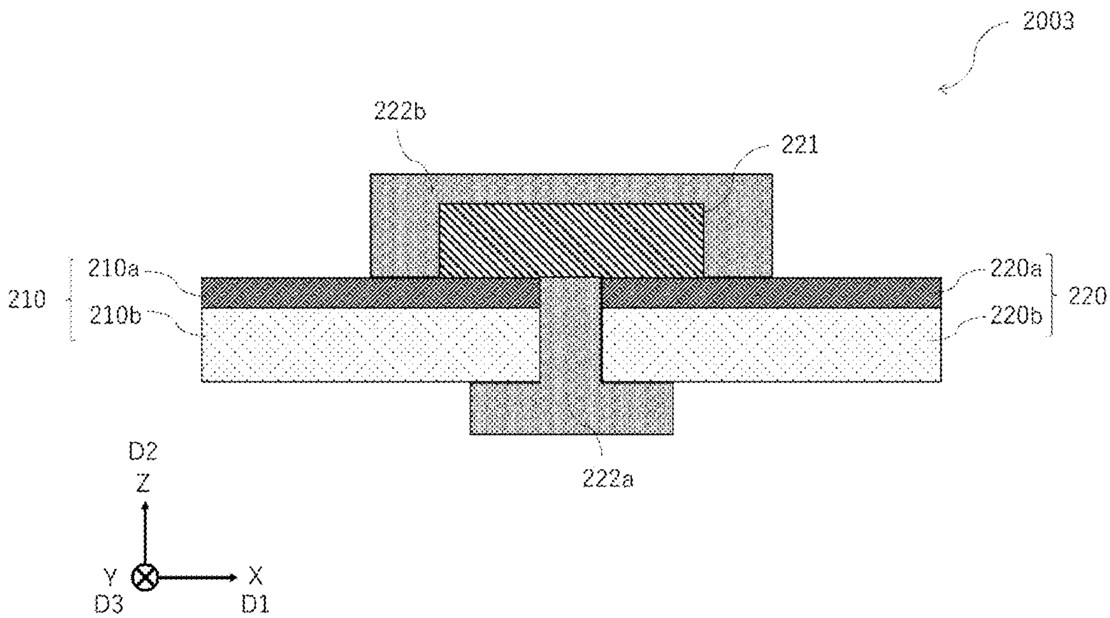

2003
222b
221
210a
210
210b
220a
220
220b
222a
D2
Z
Y
X
D3
D1

FIG. 4

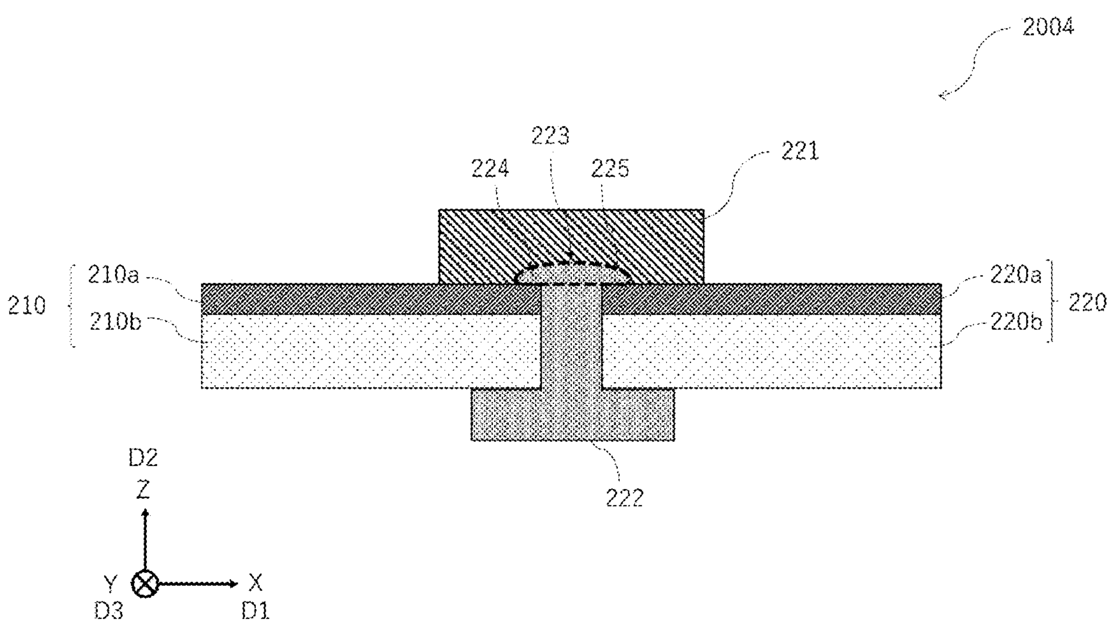

2004
223
224
225
221
210a
210
210b
220a
220
220b
222
D2
Z
Y
X
D3
D1

FIG. 5

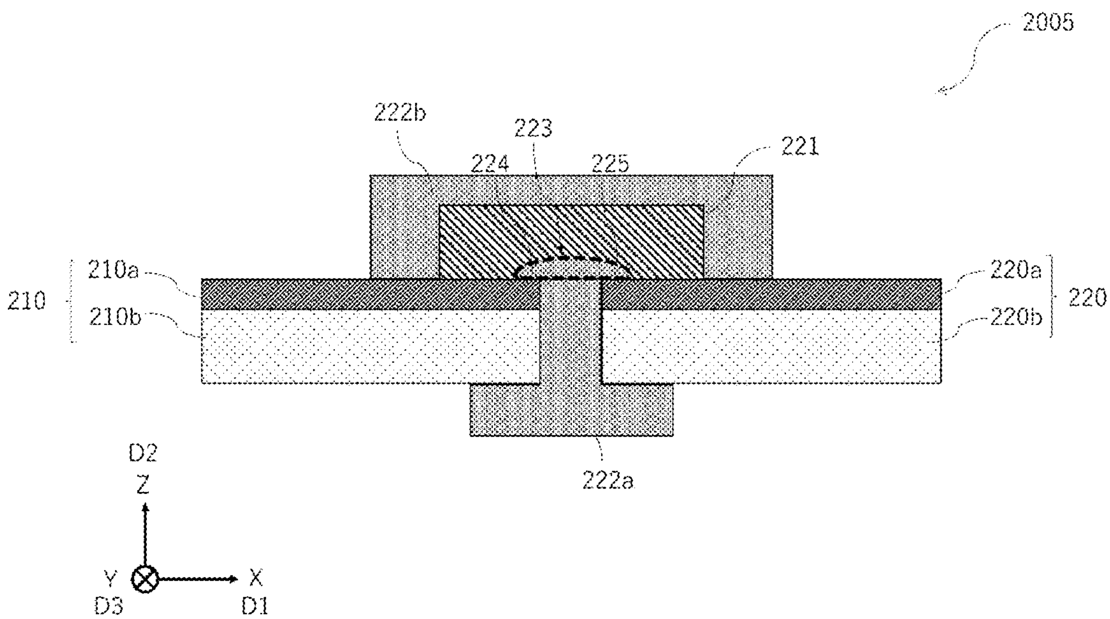

2005
222b
223
224
225
221
210a
210
210b
220a
220
220b
222a
D2
Z
Y
X
D3
D1

FIG. 6

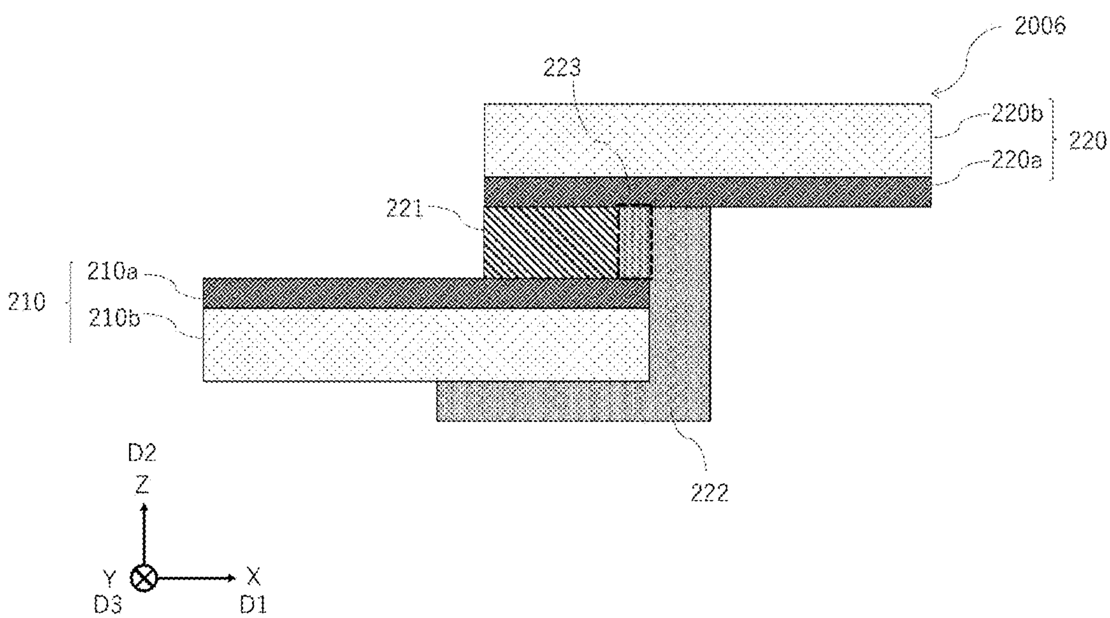

2006
223
220b
220
220a
221
210a
210
210b
222
D2
Z
Y
X
D3
D1

FIG. 7

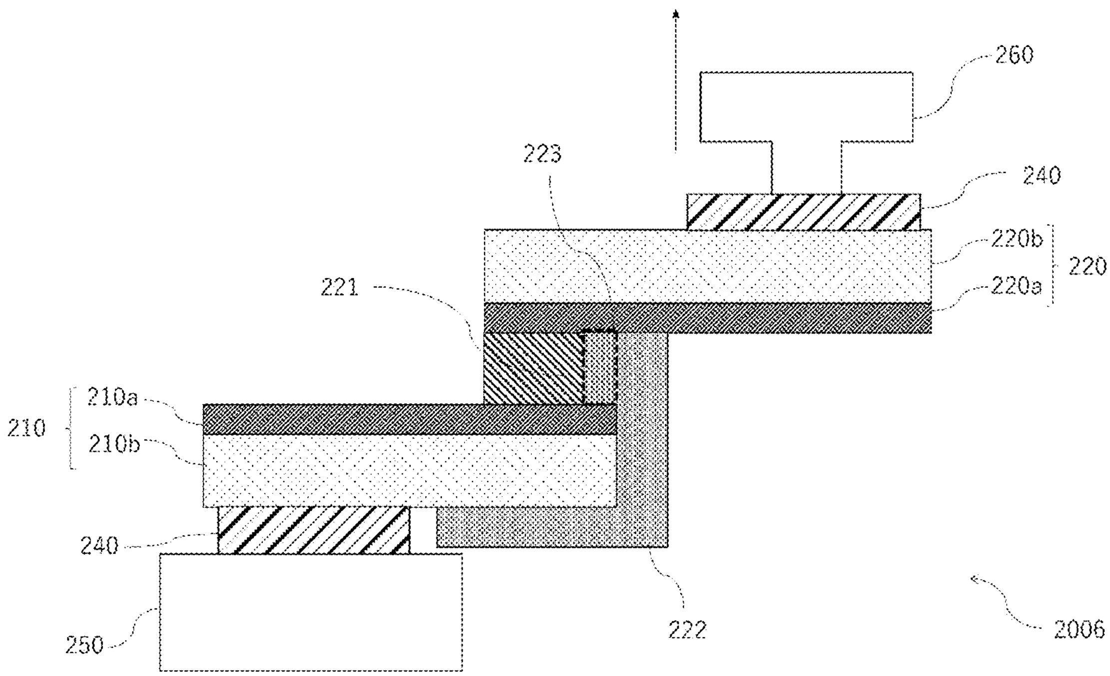

260
223
240
220b
220
220a
221
210a
210
210b
240
250
222
2006

FIG. 8

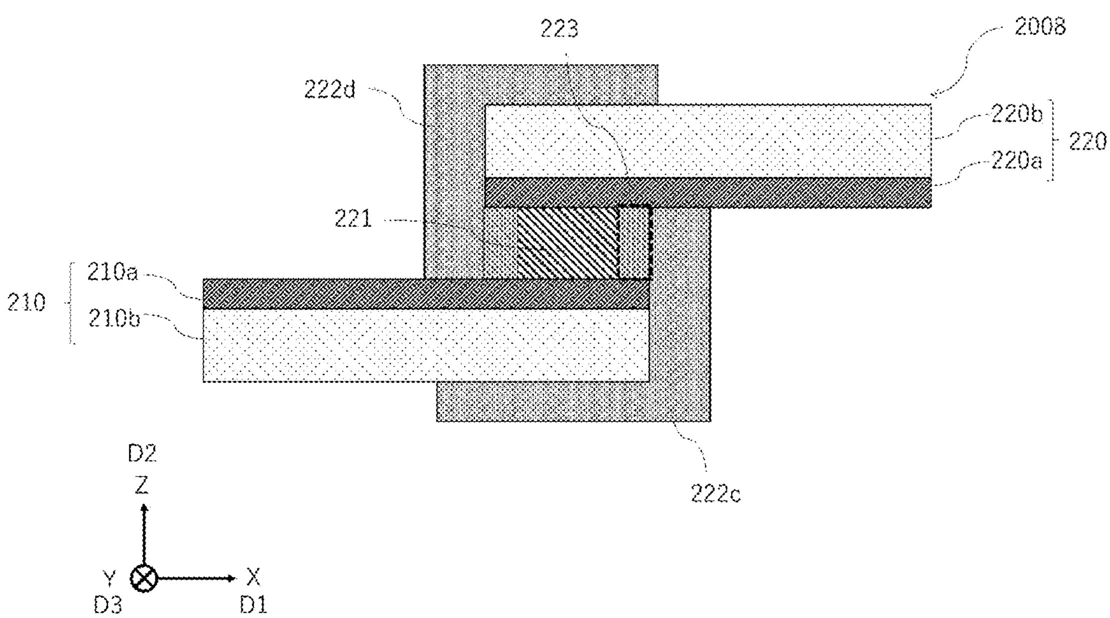

223
2008
222d
220b
220
220a
221
210a
210
210b
222c
D2
Z
Y
X
D3
D1

FIG. 9

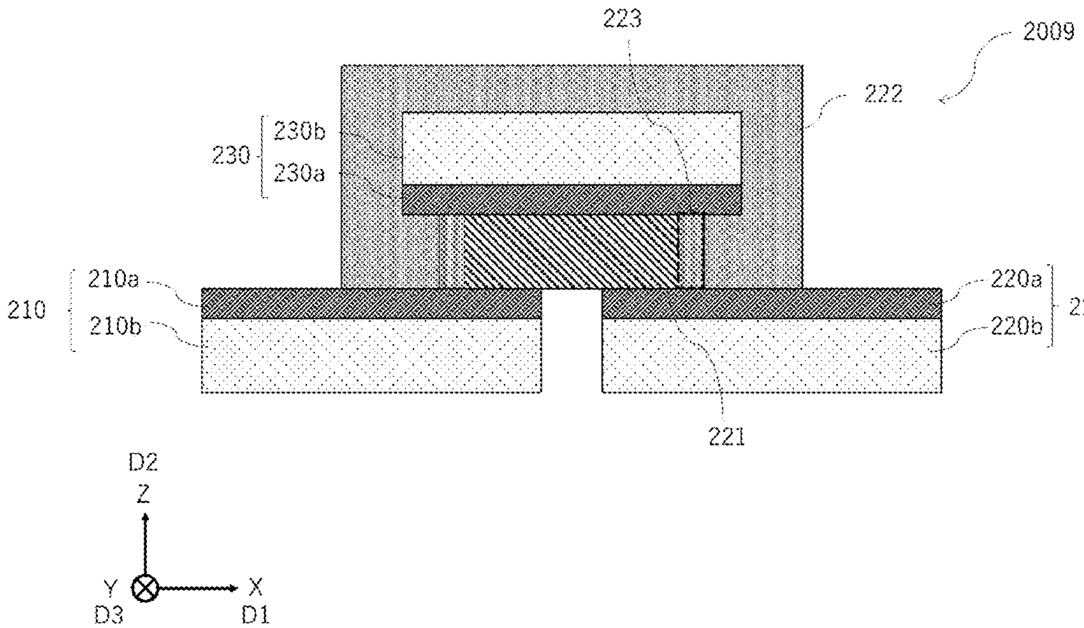

223
2009
222
230b
230
230a
210a
220a
210
220
210b
220b
221
D2
Z
Y
X
D3
D1

FIG. 10

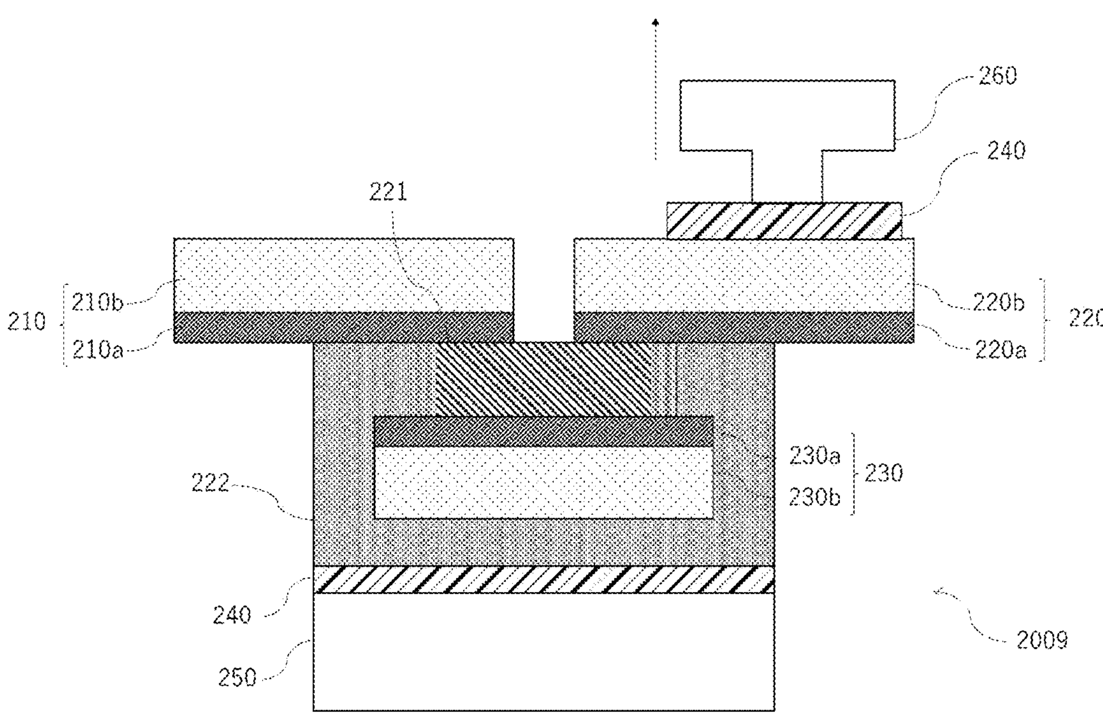

260
240
221
210b
220b
210
220
210a
220a
230a
230
230b
222
240
2009
250

FIG. 11

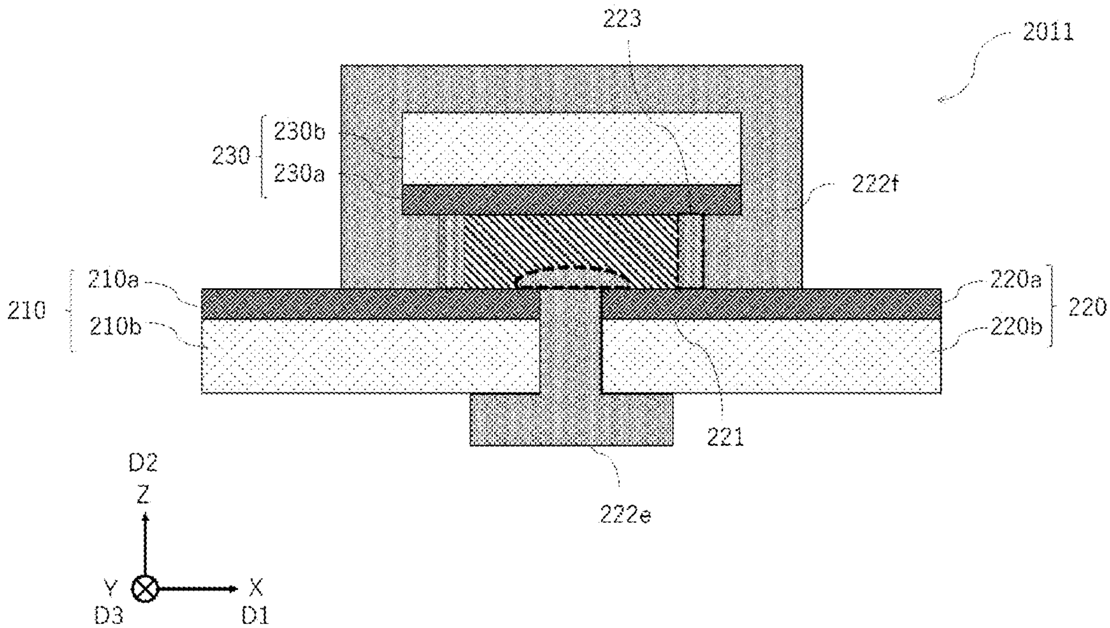

223
2011
230
230b
230a
222f
210a
210
210b
220a
220
220b
221
222e
D2
Z
Y
X
D3
D1

FIG. 12

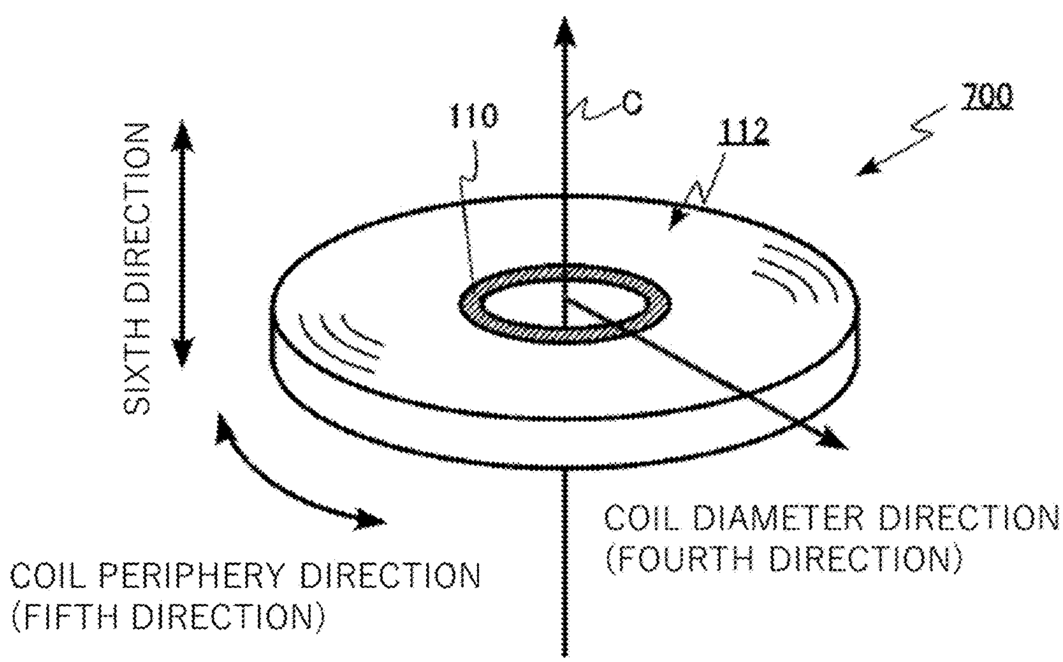

SIXTH DIRECTION
110
C
112
700
COIL DIAMETER DIRECTION
(FOURTH DIRECTION)
COIL PERIPHERY DIRECTION
(FIFTH DIRECTION)

FIG. 13

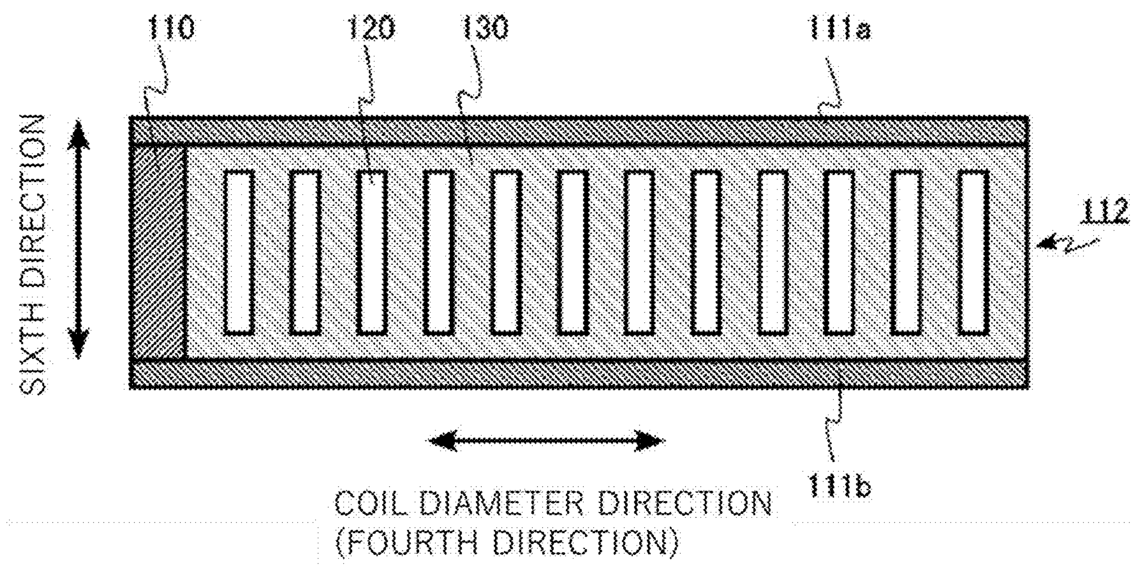
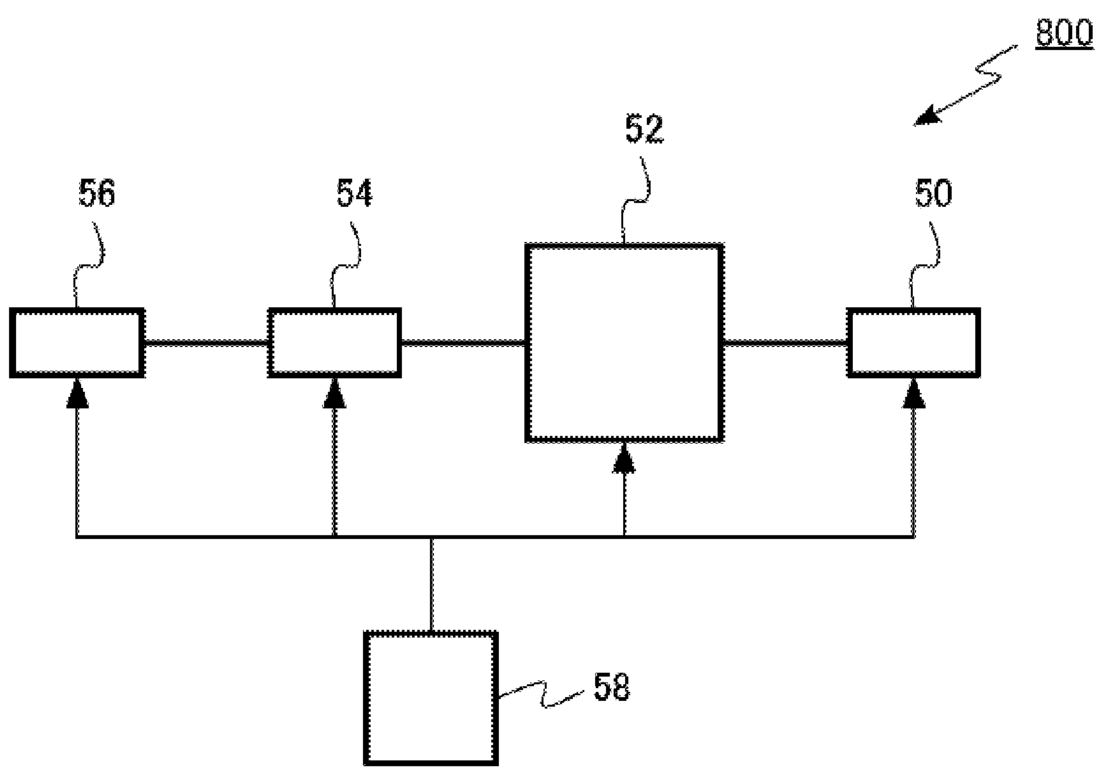

110
120
130
111a
112
SIXTH DIRECTION
111b
COIL DIAMETER DIRECTION
(FOURTH DIRECTION)

FIG. 14


800
52
56
54
50
58

SUPERCONDUCTING WIRE STRUCTURE, SUPERCONDUCTING COIL, AND SUPERCONDUCTING DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-031264, filed on Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a superconducting wire structure, a superconducting coil, and a superconducting device.

BACKGROUND

[0003] For example, in a nuclear magnetic resonance (NMR) apparatus or a magnetic resonance imaging (MRI) apparatus, a superconducting coil is used to generate a strong magnetic field. The superconducting coil is formed by winding a superconducting wire around a winding frame.

[0004] In order to lengthen the superconducting wire, for example, a plurality of superconducting wires are connected to each other. For example, the ends of two superconducting wires are connected to each other by using a connector. The connector for connecting the superconducting wires to each other is required to have a low electrical resistance and a high mechanical strength.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a schematic cross-sectional view of a superconducting wire structure according to a first embodiment;

[0006] FIG. **2** is a schematic cross-sectional view of an example of a method for measuring mechanical strength in a superconducting wire structure according to the first embodiment;

[0007] FIG. **3** is a schematic cross-sectional view of a first modification of a superconducting wire structure according to the first embodiment;

[0008] FIG. **4** is a schematic cross-sectional view of a second modification of a superconducting wire structure according to the first embodiment;

[0009] FIG. **5** is a schematic cross-sectional view of a third modification of a superconducting wire structure according to the first embodiment;

[0010] FIG. **6** is a schematic cross-sectional view of a fourth modification of a superconducting wire structure according to the first embodiment;

[0011] FIG. **7** is a schematic cross-sectional view of another example of a method for measuring mechanical strength in a superconducting wire structure according to the first embodiment;

[0012] FIG. **8** is a schematic cross-sectional view of a fifth modification of a superconducting wire structure according to the first embodiment;

[0013] FIG. **9** is a schematic cross-sectional view of a sixth modification of a superconducting wire structure according to the first embodiment;

[0014] FIG. **10** is a schematic cross-sectional view of further example of a method for measuring mechanical strength in a superconducting wire structure according to the first embodiment;

[0015] FIG. **11** is a schematic cross-sectional view of a seventh modification of a superconducting wire structure according to the first embodiment;

[0016] FIG. **12** is a schematic perspective view of a superconducting coil according to a second embodiment;

[0017] FIG. **13** is a schematic cross-sectional view of the superconducting coil according to the second embodiment; and

[0018] FIG. **14** is a block diagram of a superconducting device according to a third embodiment.

DETAILED DESCRIPTION

[0019] Hereinafter, embodiments will be described with reference to the drawings. In the following description, constituents exhibiting the same or similar function are denoted by the same reference signs throughout all drawings, and an overlapping description will be omitted.

[0020] Each drawing is a schematic view to facilitate the description and understanding of the embodiments, and the shape, size, ratio, and the like thereof may be different from those of an actual device, but these embodiments can be appropriately designed and modified by taking into consideration the following description and known techniques.

[0021] In this specification, the "particle diameter" of each particle or the like is the long diameter of the particle unless otherwise specified. The long diameter of the particle is the maximum length among the lengths between any arbitrary two points on the circumference of the particle. In addition, the short diameter of the particle is the length of a line segment that passes through the midpoint of the long diameter, is perpendicular to the long diameter, and has both ends at the circumference of the particle.

[0022] In addition, the aspect ratio of the particle is the ratio of the long diameter to the short diameter (long diameter/short diameter) of the particle. The long and short diameters of the particle can be calculated, for example, by image analysis of scanning electron microscope (SEM) images.

First Embodiment

[0023] A superconducting wire structure according to a first embodiment includes: a first superconducting wire; a second superconducting wire adjacent to the first superconducting wire; a joint layer electrically connecting the first superconducting wire and the second superconducting wire; and a connecting member connecting the joint layer and at least one of the first superconducting wire and the second superconducting wire. The connecting member has a melting point equal to or more than 900° C. and equal to or less than 1100° C.

[0024] FIG. **1** is a schematic cross-sectional view of a superconducting wire structure **200** according to the first embodiment. A superconducting wire structure **200** according to the first embodiment, for example, is used to lengthen superconducting wires by joining two superconducting wires (a first wire **210** and a second wire **220**) to each other.

[0025] Here, a first direction D**1** crossing a direction from the first wire **210** to the joint layer **221** and along the first wire **210** is defined as an X-axis direction. A second direction D**2** from the first wire **210** to the joint layer **221** is defined as a Z-axis direction. One direction perpendicular to the Z-axis direction is defined as an X-axis direction. A direction perpendicular to the Z-axis direction and the X-axis direction is defined as a Y-axis direction.

[0026] The superconducting wire structure **200** includes the first wire **210**, the second wire **220**, the joint layer **221**, and a connecting member **222**.

[0027] The connecting member **222** connects the joint layer **221** and at least one of the first wire **210** and the second wire **220**.

[0028] In FIG. **1**, the connecting member **222** is in contact with a part of a lower surface of the joint layer **221** having an interface with a part of the first wire **210** and a part of the second wire **220**. Thus, the joint layer **221**, the first wire **210**, and the second wire **220** are physically connected through the connecting member **222**. The connecting member **222** is preferably provided to connect the joint layer **221**, the first wire **210**, and the second wire **220** because it is possible for mechanical strength of the superconducting wire structure **200** to be improved.

[0029] The connecting member **222** is, for example, metal. The connecting member **222** includes at least one element selected from the group composing of Ag, Cu, Au, and Ge.

[0030] The connecting member **222** has a melting point equal to or more than 900° C. and equal to or less than 1100° C. In manufacturing the superconducting wire structure **200** according to the embodiment, a heat treatment with the superconducting wire structure **200** pressurized is, for example, equal to or more than 700° C. and equal to or less than 850° C. Thus, the temperature of the heat treatment is equal to or more than 70% of the melting point of the connecting member **222**. Hence, it is possible for the connecting member **222** to be easily softened during the heat treatment.

[0031] Since the melting point of the connecting member **222** is equal to or more than 900° C., it becomes possible to prevent particles composing the joint layer **221** from being extruded when the connecting member **222** melts during a heat treatment of the superconducting member **200** with the superconducting wire structure **200** being pressurized. Therefore, it becomes possible to prevent formation of a region where particles do not exist in the joint layer **221**. Thus, it becomes possible for a sufficient path for current in the joint layer **221** to be obtained. Furthermore, in the joint layer **221** according to the embodiment, the heat treatment is performed while the particles are in contact with each other, causing a reaction between the particles which makes it possible to obtain a sufficient path for current in the joint layer **221**. Accordingly, raising the temperature of the heat treatment over the melting point of the connecting member **222** makes it possible to prevent a portion of the melting connecting member **222** from entering between the particles in the joint layer **221** before the particles react in the joint layer **221**.

[0032] In addition, since the melting point of the connecting member **222** is equal to or less than 1100° C., it becomes possible for the temperature of the heat treatment to be equal to or more than 70% of the melting point of the connecting member **222**. Thus, it is possible for the connecting member **222** to contribute to the connection of each wire **210** and **220**.

[0033] Hereinafter, another aspect of the superconducting wire structure **220** according to an embodiment will be described.

(First Wire)

[0034] The first wire **210** includes a first conducting layer **210***a*, and a first substrate **210***b*. The first conducting layer **210***a* is provided on the first substrate **210***b*.

[0035] The first wire **210** is electrically connected to the second wire **220** through the joint layer **221**. Therefore, it is possible for the current to flow between the first wire **210** and the second wire **220** through the joint layer **221**.

[0036] The first substrate **210***b* is, for example, a metal. The first substrate **210***b* is, for example, a nickel alloy or a copper alloy.

[0037] The first substrate **210***b* is, for a specific example, a nickel-chromium-molybdenum alloy.

[0038] The first superconducting layer **210***a* is, for example, an oxide superconducting layer. The first superconducting layer **210***a* contains, for example, a rare earth element (RE), barium (Ba), copper (Cu), and oxygen (O). For example, the first superconducting layer **210***a* contains at least one rare earth element (RE) in a group composed of: yttrium (Y), lanthanum (La), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), and lutetium (Lu).

[0039] The first superconducting layer **210***a* has, for example, a chemical composition represented by (RE) $Ba_2Cu_3O_\delta$ (RE is a rare earth element, $6\leq\delta\leq7$). Specifically, the first superconducting layer **210***a* has a chemical composition represented by, for example, $GdBa_2Cu_3O_\delta$ ($6\leq\delta\leq7$), $YBa_2Cu_3O_\delta$ ($6\leq\delta\leq7$), or $EuBa_2Cu_3O_\delta$ ($6\leq\delta\leq7$). The first superconducting layer **210***a* contains, for example, a rare earth oxide superconductor.

[0040] The first superconducting layer **210***a* contains, for example, a single crystal having a perovskite structure.

(Second Wire)

[0041] The second wire **220** is provided adjacent to the first superconducting wire **210**.

[0042] The second wire **220** includes a second superconducting layer **220***a*, and a second substrate **220***b*. The second superconducting layer **220***a* is provided on the second substrate **220***b*.

[0043] The second substrate **220***b* is, for example, a metal as the first substrate **210***b*. The second substrate **220***b* may include the same material as the first substrate **210***b* or different materials.

[0044] The second superconducting layer **220***a* is, for example, an oxide superconducting layer similar as the first superconducting layer **210***a*. The second superconducting layer **220***a* may include the same material as the first substrate **210***a* or different materials.

(Joint Layer)

[0045] The joint layer **221** is in a contact with at least a part of the first wire **210** and at least a part of the second wire **220**. The joint layer **221** electrically connects the first wire **210** and the second wire **220**.

[0046] The joint layer **221** is connected to one of the first wire **210** and the second wire **220** through the connecting member **222**. For example, the connecting member **222**

connects a side surface of the first wire **210** crossing a first direction from the first wire **210** to the second wire **220** with a lower surface of the joint layer **221** having an interface with the first wire **210**. Here, for example, the first direction is the X-axis direction. Since the connecting member **222** connects the exposed side surface of the first wire **210** with the exposed lower surface of the joint layer **221**, it is possible to prevent peeling of the particles composing the joint layer **221** from the first wire **210** caused by entry of the connecting member **222** to the joint layer **221**. Therefore, it is possible to prevent deterioration of the conduction path.

[0047] The joint layer **221** contains, for example, single crystal particle groups containing a rare earth element (RE), barium (Ba), copper (Cu), and oxygen (O) or polycrystal particle groups containing the same elements as the single crystal. The joint layer **221** is, for example, composed of a particle group of a single crystal containing at least one rare earth element (RE) in a group composing of yttrium (Y), lanthanum (La), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), and lutetium (Lu) or polycrystal containing the same elements as the single crystal.

[0048] The joint layer **221** is, for example, composed of a particle group of a single crystal having a chemical composition represented by (RE) $Ba_2Cu_3O_\delta$ (RE is a rare earth element, $6 \le \delta \le 7$) or polycrystal having the same chemical composition as the single crystal. Specifically, the joint layer **221** is composed of a particle group of a single crystal having a chemical composition represented by, for example, $GdBa_2Cu_3O_\delta$ ($6 \le \delta \le 7$), $YBa_2Cu_3O_\delta$ ($6 \le \delta \le 7$), or $EuBa_2Cu_3O_\delta$ ($6 \le \delta \le 7$) or polycrystal having the same chemical composition as the single crystal. The joint layer **221** is, for example, composed of a particle group of a single crystal of a rare earth oxide superconductor or polycrystal of the rare earth oxide superconductor.

[0049] The average particle size of particles in these particle groups is equal to or more than 500 nm and equal to or less than 30 μm, preferably equal to or more than 1 μm and equal to or less than 10 μm. When the average particle size of particles in these particle groups is equal to and more than 500 nm, sufficient crystallinity can be obtained, and thus, high superconducting properties can be obtained. Further, when the particle size is equal to and less than 30 μm, the thickness of the joint layer **221** can be controlled so as not to be increased. This can reduce portions that become resistance portions such as voids existing in joint layer **221**, and thus, the resistance of the superconducting wire structure **200** can be reduced. In addition, when the particle size is equal to or less than 30 μm, the contact area between the joint layer **221** and the first wire **210** and the contact area between the joint layer **221** and the second wire **220** can be sufficiently increased, and thus, the mechanical strength of the superconducting wire structure **200** can be increased.

[0050] The aspect ratio of these particle groups is equal to or more than 1.5 and equal to or less than 50, preferably equal to or more than 3 and equal to or less than 10. When the aspect ratio is equal to or more than 1.5, the joint layer **221** can have a characteristic as superconductivity. In addition, when the aspect ratio is equal to or less than 50, the bulk density of the joint layer **221** can be sufficiently increased, and the thickness of the joint layer **221** can be controlled so as not to be increased. Furthermore, the contact area between the joint layer **221** and the first wire **210** and the contact area between the joint layer **221** and the second wire **220** can be sufficiently increased.

[0051] The porosity of the joint layer **221** is equal to or more than 10% and equal to or less than 70%, preferably equal to or more than 20% and equal to or less than 50%. When the porosity is equal to or more than 10%, a part of the connecting member **222** easily enters the region that exists in the vicinity of the interfacial surface between the first superconducting wire **210** and the joint layer **221**, so that the mechanical strength can be increased. When the porosity is equal to or less than 70%, the amount of particles in the joint layer **221** can be made sufficient, so that a decrease in critical current due to a shortage of current paths can be prevented.

[0052] Here, a method for manufacturing the superconducting wire structure according to the embodiment in FIG. **1** will be described.

[0053] First, the first wire **210** and the second wire **220** are prepared. Here, when each superconducting layer has a protection layer (mainly a Cu layer and an Ag layer) for preventing the superconducting layer from reacting with moisture in the atmosphere, for example, this layer is removed by using a wet etching method, and each of the superconducting layer **210*a*** and **220*a*** is exposed.

[0054] Thereafter, the superconducting layers of the first wire **210** and the second wire **220** are configured to be oriented in the same direction. In addition, the connecting member **222** is arranged so that at least a part of each wire is in contact with the connecting member **222**. Here, the connecting member **222** is obtained by using, for example, a metal sheet. The connecting member **222** to be used is preferably equal to or more than 0.001 mm and equal to or less than 1 mm in thickness. When the thickness of the connecting member **222** is equal to or more than 0.001 mm, sufficient strength of superconducting wire structure **200** can be obtained. In addition, since the thickness is equal to or less than the 1 mm, heat generated during the heat treatment sufficiently reaches the inside of the connecting member **222**, and the connecting member **222** can be easily softened. Therefore, the first wire **210** and the second wire **220** can be connected to each other by the connecting member **222**.

[0055] Then, the single crystal or polycrystal particles of the rare earth oxide superconductor described in the section above of (Joint Layer) and a solution containing an organic compound containing the same kind of metal element as a main component are mixed at a ratio of 5:1 to 1:3 depending on the average particle diameter of the particles, to obtain a mixture for the joint layer **221**. The mixture obtained as described above is molded on the first wire **210** and the second wire **220** to have a necessary thickness. Then, drying is performed, and baking is performed in oxygen at equal to or more than 700° C. and equal to or less than 850° C., so that the joint layer **221** is formed on the first wire **210** and the second wire **220**.

[0056] In this way, a structure for obtaining the superconducting wire structure **200** in FIG. **1** is obtained.

[0057] Next, in the structure obtained as described above, a weight is placed on a portion where the joint layer **221** and each wire **210** and **220** are overlapped with each other to apply pressure. For example, a jig for pressurization is prepared and the structure is sandwiched between the jig to apply pressure. The jig may be removed after the connection, or the jig may be left attached. The jig is preferably removed because the superconducting wire structure **200** can be easily wound around a coil.

[0058] Next, heat treatment is performed in a pressurized state. The heat treatment is performed, for example, in argon containing O_2 having concentration of equal to or more than 100 ppm and equal to or less than 1500 ppm at a temperature of equal to or more than 700° C. and equal to or less than 850° C., and then, in O_2 at a temperature of equal to or more than 300° C. equal to or less than to 600° C.

[0059] The connecting member **222** is softened by the heat treatment, and the connecting member **222** as shown in FIG. **1** is obtained.

[0060] In this way, superconducting wire structure **200** in FIG. **1** is obtained.

[0061] The critical current of the superconducting wire structure **200** according to the present embodiment can be measured, for example, as follows.

<Measurement of Critical Current>

[0062] The critical current is the maximum current value that can be passed at zero electrical resistance, and it can be measured by a four terminal measurement method. The terminal is attached to the superconducting wire structure **200** after the connection, and measurement is performed in liquid nitrogen. Since the voltage increases as the resistance increases, the current value at which the voltage starts to increase is defined as the critical current.

[0063] The mechanical strength of the superconducting wire structure according to the present embodiment can be evaluated, for example, according to Japanese Industrial Standards JIS K 6849 (1994) "Testing methods for tensile strength of adhesive bonds (method of measuring by tensile load perpendicular to the bonding surface)". Each specific measurement method in the example of FIG. **1** will be described with reference to FIG. **2**. FIG. **2** is a schematic cross-sectional view of an example of a method for measuring mechanical strength in the example of the superconducting wire structure shown in FIG. **1**.

<Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>

(Fixing of Superconducting Wire Structure)

[0064] In FIG. **2**, one surface of a double-sided adhesive tape **240** is bonded to the first substrate **210***b* or the second substrate **220***b* (in the example shown to the substrate **210***b*). In addition, the other surface of the double-sided adhesive tape **240** is adhered to a pedestal **250** and the superconducting wire structure **200** structure is fixed to the pedestal **250**. At this time, the double-sided adhesive tape **240** is prepared so that the width of the tape **240** is the same length as the wire for fixing the pedestal **250** and the substrate of one of the wires. Further, the length of the tape **240** is a length that enables strong enough adhesion that does not cause peeling between the pedestal **250** and the substrate during measurement of the mechanical strength.

[0065] Next, one surface of another double-sided adhesive tape **240** is bonded to at least a part of the superconducting layer of the wire that is not fixed to the pedestal **250**. In the case of FIG. **2**, the superconducting layer of the wire is the second superconducting layer **220***a* of the second wire **220**. At this time, the double-sided adhesive tape **240** is prepared so that the width of the tape **240** is the same length as the wire for fixing the superconducting layer of the other wire and the tensile device **260**. Further, the length of the tape **240** is a length that enables strong enough adhesion that does not cause peeling between the pedestal **250** and the substrate during measurement of the mechanical strength. In addition, the double-sided adhesive tape **240** used for fixing the tensile device **260** and the wire is provided at a certain distance from the end of the joint layer **221**. For example, the distance can be equal to or more than 1 mm and equal to or less than 40 mm, preferably equal to or more than 5 mm and equal to or less than 20 mm.

[0066] The double-sided adhesive tape **240** may be any tape as long as it has a sufficient adhesive strength to withstand the measurement, and for example, a carbon tape for SEM observation can be used.

[0067] When the mechanical strength of the superconducting wire structure for the example of FIG. **2** is measured, the double-sided adhesive tape **240** having the above-described size is used for each measurement.

(Measurement Using Tensile Device)

[0068] As the operation of the tensile device **260**, the tensile device **260** is operated in a direction away (the direction of the arrow in FIG. **2**) from the wire (the second wire **220** in FIG. **2**) fixed to the tensile device **260** via the double-sided adhesive tape **240**. At this time, the operation speed of the tensile device **260** is set to be constant, for example, equal to or more than 100 μm/s and equal to or less than 5 mm/s, preferably equal to or more than 500 μm/s and equal to or less than 1 mm/s. Since the operation of the tensile device **260** is equal to or more than 100 μm/s and equal to or less than 5 mm/s, the measurement time can be shortened within a range in which measurement errors due to the wires being immediately peeled off are suppressed. As the measurement, for example, the distance by which the tensile device **260** moves until the wire is completely peeled off from the joint layer **221** can be measured as the mechanical strength.

[0069] The mechanical strength between the first wire **210** or the second wire **220** and the connection layer **221** can be evaluated by the above-described method.

[0070] The porosity of joint layer **221** of superconducting wire structure **200** can be measured as follows after the above-described mechanical strength test.

<Method for Measuring Porosity of Joint Layer>

[0071] First, the connection portion including the joint layer **221** is embedded in, for example, an epoxy-based resin, and the vicinity of the center of the connection portion which is firmly connected is cut by a diamond saw or the like because peeling is a concern at the end of the connection. At this time, in a modification example described later where the connecting member **222** is, for example, disposed between the joint layer **221** and the first wire **210**, the cross section for measuring the porosity of the joint layer **221** is located at a position avoiding the connecting member **222**. In addition, in a modification example described later where the connecting member **222** has entered the joint layer **221**, the porosity of the joint layer **221** is measured after the connecting member **222** is measured.

[0072] Next, the cut surface is polished with waterproof paper and buffed or the like, and then coated with a conductive material, and the observation positions are determined by SEM. Ion milling is performed on the determined observation positions, and SEM observation is performed after conductive coating is performed.

[0073] Although depending on the average particle diameter of the powder used, when the average particle diameter is equal to or more than 2 μm and equal to or less than to 3 μm, SEM images of three positions in the cross section are taken at a magnification of about 2500 times. At this time, the joint layer **221** is processed to have three cross sections at positions where the sides corresponding to the long sides of the substantially quadrangular shape of the joint layer **221** in plan view are equally divided into four. The magnification of the SEM image is set such that the larger the average particle diameter, the lower the magnification, and the smaller the average particle diameter, the higher the magnification. The SEM image is binarized to separate particles from voids, for example, using ImageJ, and then the areas of the particles and the voids are calculated. The porosity in each cross section is calculated from the ratio of the areas. The porosity obtained in each cross section is averaged to obtain the porosity of the joint layer **221** to be obtained.

[0074] Hereinafter, a modification of the superconducting wire structure according to the first embodiment will be described.

[0075] FIG. **3** is a schematic cross-sectional view of a first modification of the superconducting wire structure according to the first embodiment.

[0076] In FIG. **3**, the connecting member includes a connecting member **222***a* positioned between the first wire **210** and the second wire **220**. In FIG. **3**, the connecting member further includes a connecting member **222***b* covering a part, except for the interfaces between the first wire **210** and the second wire **220**, exposed to the outside of the joint layer **221**.

[0077] The connecting member **222***b* is in contact with at least a part of the first superconducting layer **210***a* and at least a part of the second superconducting layer **220***a*. This improves the mechanical strength between the joint layer **221** and the first wire **210** and the mechanical strength between the joint layer **221** and the second wire **220**, and thus the mechanical strength of the superconducting wire structure **2003** can be improved.

[0078] When a surface of the joint layer **221** facing the lower surface having the interface with each wire **210** and **220** is defined as an upper surface and a surface of the joint layer **221** located between the upper surface and the lower surface of the joint layer **221** is defined as a side surface, the connecting member **222***b* is not necessarily in contact with the entire upper surface and the entire side surfaces.

[0079] The connecting member **222***b* may be provided such that the joint layer **221** is in contact with either or both the first wire **210** and the second wire **220**. For example, there may be a region where a part of the side surface of the joint layer **221** is not covered. In addition, a microscopic gap may be present between the connecting member **222***b* and the joint layer **221**. The connecting member **222***b* is preferably in contact with substantially the entire upper surface and the entire side surfaces of the joint layer **221**.

[0080] The superconducting wire structure **2003** in FIG. **3** can be manufactured as follows. Here, the description of the same parts as those of the method of manufacturing the superconducting wire structure **200** in FIG. **1** will be omitted.

[0081] First, after the above-described method for manufacturing a structure for obtaining the superconducting wire structure **200** of FIG. **1** is performed, the connecting member **222***b* is disposed to be in contact with the joint layer **221**. Thereafter, the heat treatment with the above-described structure being pressurized is performed to obtain the superconducting wire structure **2003** in FIG. **3**.

[0082] The critical current, the mechanical strength, and the porosity of the joint layer **221** of the obtained superconducting wire structure **2003** of FIG. **3** can be measured in the same manner as the methods described above in <Measurement of Critical Current>, <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and <Method for Measuring Porosity of Joint Layer>.

[0083] FIG. **4** is a schematic cross-sectional view of a second modification of the superconducting wire structure according to the first embodiment.

[0084] In this further embodiment the connecting member **222** is also present inside the joint layer **221** in FIG. **4**. The region **223** is a portion where a part of the connecting member **222** exists inside the joint layer **221**. The region **223** is generated by, for example, a part of the connecting member **222** entering the joint layer **221**. This is because the joint layer **221** is formed of the particle groups, and the region **223** is generated by a part of the connecting member **222** entering a void area or areas present between the particle groups. This can improve the mechanical strength of superconducting wire structure **2004**, and thus the region **223** is preferably present.

[0085] The connecting member **222** covers a part of the surface of the first substrate **210***b* and a part of the surface of the second substrate **220***b*

[0086] The superconducting wire structure **2004** in FIG. **4** can be manufactured in the same manner as the method for fabricating superconducting wire structure **200** in FIG. **1** described above.

[0087] In FIG. **4**, the connecting member **222** is softened by the heat treatment, whereby the connecting member **222** enters the joint layer **221** and the region **223** is formed.

[0088] The region **223** includes at least one of a first region **224** present between the joint layer **221** and the first wire **210**, and a second region **225** present between the joint layer **221** and the second wire **220**. In this case, the first region **224** and the second region **225** satisfy at least one of a first condition and a second condition. In the first condition, the first region **224** is located in a range of equal to or more than 5 μm and equal to or less than 1 mm from an end of the first wire **210** in the first direction D**1**. The second condition is that the second region **225** is present in a range of equal to or more than 5 μm and equal to or less than 1 mm from an end of the second wire **220** in the first direction D**1**. Preferably, the first condition and the second condition are satisfied. That is, it is preferable that the first region **224** is present in a range of equal to or more than 5 μm and equal to or less than 1 mm from the end of the first wire **210**, and the second region **225** is present in a range of equal to or more than 5 μm and equal to or less than 1 mm from the end of the second wire **220**. These can improve the mechanical strength of the superconducting wire structure **2004**.

[0089] Preferably, at least one of the first condition and the second condition is satisfied, and the connecting member **222** existing in the first region **224** and the first wire **210** are in contact with each other, or the connecting member **222**

existing in the second region **225** and the second wire **220** are in contact with each other. Thereby, the mechanical strength of the superconducting wire structure **2004** can be improved.

[0090] The region **223** is preferably present in the range of equal to or less than 1 mm from the interface between the first wire **210** or the second wire **220** and the joint layer **221** also in the second direction D**2**. This can improve the mechanical strength of the superconducting wire structure **2004**.

[0091] The upper limit of the range from the end of the first wire **210** or the end of the second wire **220** where the first region **224** or the second region **225** may be present is preferably equal to or less than 200 μm. Since the first region **224** or the second region **225** is present at a position equal to or more than 5 μm from the end of the first wire **210** or the end of the second wire **220**, the mechanical strength as the superconducting wire structure **2004** can be increased. In addition, when the first region **224** or the second region **225** is present in the range of equal to or less than 1 mm from the end of the first wire **210** or the end of the second wire **220**, the intrusion of the connecting member **222** in the joint layer **221** by pushing the particles composing of the joint layer **221** in the first direction can be suppressed. This makes it possible to obtain a sufficient connection area and to provide a sufficient current path, thereby it is possible to prevent a decrease in critical current. Here, the connection area is a contact area between the first wire **210** or the second wire **220** and the joint layer **221** when a current passes between the wires via the joint layer **221**.

[0092] Therefore, to further increase the mechanical strength of superconducting wire structure **2004** and to achieve a higher critical current by obtaining sufficient connection areas, the range from the end of first wire **210** or the end of second wire **220** where first region **224** or second region **225** can exist is preferably equal to or more than 5 μm and equal to or less than 1 mm.

[0093] The critical current, the mechanical strength, and the porosity of the joint layer **221** of the obtained superconducting wire structure **2004** of FIG. **4** can be measured in the same manner as the methods described above in <Measurement of Critical Current>, <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and <Method for Measuring Porosity of Joint Layer>. However, in the case where the length of the connecting member **222** along the first direction in which the connecting member **222** enters the joint layer **221** is measured, the measurement is performed before the measurement of the porosity of the joint layer **221**.

[0094] Here, a method for measuring the length of the connecting member **222** entering the joint layer **221** along the first direction will be described.

<Method for Measuring Length Along First Direction of Member Entering Joint Layer>

[0095] When the connecting member **222** includes the first region **224** as shown in FIG. **4**, the distance can be measured by measuring the connecting member **222** remaining on the first superconducting layer **210***a* of the first wire **210** after the separation of the joint layer **221** and, for example, the first wire **210**.

[0096] First, after <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**> is performed, the joint layer **221** and, for example, the first wire **210** are separated from each other.

[0097] Thereafter, a mapping image of elements derived from the components of the connecting member **222** is captured by EDX (Energy Dispersive X-ray Spectroscope) on the surfaces of the first superconducting layer **210***a* of the first wire **210**. This makes it possible to measure the depth of the connecting member **222** entering the joint layer **221** in the first direction D**1**.

[0098] When capturing EDX images, first, the area on the surface of the first superconducting layer **210***a*, where the joint layer **221** existed before separation from the first wire **210**, is divided into five equal parts along the third direction D**3**, which intersects with the first direction D**1** and is parallel to the surface. The third direction D**3** is, for example, the Y-axis direction along the surface of the first superconducting layer **210***a*, intersecting with the X-axis. The regions at both ends of the five equal parts along the Y-axis direction are excluded from the EDX imaging area. That is, the central three regions of the five equal parts along the third direction D**3** are set as the EDX imaging area. The EDX imaging magnification is set to a range where the connecting member **222** with the longest distance along the first direction D**1** is within the field of view in each imaging area. In each EDX image obtained in this way, the average length of the connecting member **222** in the third direction on the surface of the first superconducting layer **210***a* is determined. The average length of the invasion mentioned above is determined, for example, at positions dividing each width of the EDX images along the third direction D**3** into six equal parts. By averaging these five lengths obtained from each of the three EDX images, a total of 15 lengths, the length of the connecting member **222** invading the joint layer **221** along the first direction D**1** can be determined.

[0099] FIG. **5** is a schematic cross-sectional view of a third modification of the superconducting wire structure according to the first embodiment.

[0100] In FIG. **5**, the connecting member **222** has a second connecting member **222***b* covering a portion exposed outside the joint layer **221**, excluding the interface with the first wire **210** and the second wire **220** in addition to the first connecting member **222***a* located between the first wire **210** and the second wire **220**.

[0101] The second connecting member **222***b* also may be present inside the joint layer **221**, and the region **223** may exist in FIG. **5**.

[0102] The superconducting wire structure **2005** in FIG. **5** can be manufactured in the same manner as the method of fabricating the superconducting wire structure **2003** in FIG. **3** described above.

[0103] The critical current, the mechanical strength, and the porosity of the joint layer **221** of the obtained superconducting wire structure **2005** of FIG. **5** can be measured in the same manner as the methods described in <Measurement of Critical Current>, <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and <Method for Measuring Porosity of Joint Layer>.

[0104] In FIG. **5**, the second connecting member **222***b* may include the region **223** by the second member **222***b* entering the joint layer **221** along the second direction D**2**. The region is preferably present in a range of equal to or more than 5 μm and equal to or less than 1 mm from the surface of the joint layer **221** in the second direction D**2**. The region can be identified by observing a cross section of the joint layer **221** in a plane intersecting with the X-Y plane formed by the X axis and the Y axis, for example, in the Z-X plane.

[0105] FIG. **6** is a schematic cross-sectional view of a fourth modification of the superconducting wire structure according to the first embodiment.

[0106] In FIG. **6**, the first superconducting layer **210***a* and the second superconducting layer **220***a* are provided to face each other, and the joint layer **221** is provided to be positioned between the first superconducting layer **210***a* and the second superconducting layer **220***a*.

[0107] In this case, the superconducting layers of the first wire **210** and the second wire **220** face each other and the joint layer **221** is interposed therebetween, and thus the joint layer **221** is protected. For example, when the superconducting wire structure **2006** of FIG. **6** is wound into a coil, the joint layer **221** of the superconducting wire structure **2006** wound in the n-th turn is protected from contact with the superconducting wire structure **2006** in the (n+1)-th turn, and thus wear of the joint layer **221** can be suppressed.

[0108] The connecting member **222** connects the joint layer **221** to at least one of the first wire **210** and the second wire **220**. In FIG. **6**, the connecting member **222** is in contact with one surface of the joint layer **221**.

[0109] The connecting member **222** enters the joint layer **221** and forms a region **223** that exists in a range of equal to or more than 5 μm and equal to or less than 1 mm from the end of the first wire **210**.

[0110] The superconducting wire structure **2006** in FIG. **6** can be manufactured as follows.

[0111] Firstly, at least a part of the first substrate **210***b* and the connecting member **222** are disposed to be in contact with each other. Thereafter, the joint layer **221** is applied on at least a part of the second superconducting layer **220***a*. Secondly, at least a part of the first superconducting layer **210***a*, and in the joint layer **221** at least a part of a surface on the opposite side to the surface with which the second superconducting layer **220***a* is in contact, are provided to face each other. That is, the joint layer **221** is located between the first superconducting layer **210***a* and the second superconducting layer **220***a*.

[0112] Thus, a structure for obtaining the superconducting wire structure **2006** in FIG. **6** is obtained. Thereafter, the superconducting wire structure **2006** in FIG. **6** is obtained by the above-described pressurizing method and heat treatment.

[0113] The critical current, the mechanical strength, and the porosity of the joint layer **221** of the obtained superconducting wire structure **2006** of FIG. **6** can be measured in the same manner as the methods described above in <Measurement of Critical Current>, <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and <Method for Measuring Porosity of Joint Layer>.

[0114] A method for measuring the mechanical strength in FIG. **6** will be described with reference to FIG. **7**. Here, FIG. **7** is a schematic cross-sectional view showing a method of measuring the mechanical strength in the example of the superconducting wire structure **2006** shown in FIG. **6**.

<Method for Measuring Mechanical Strength of Superconducting Wire in FIG. **6**>

[0115] In FIG. **7**, the substrate of one wire and the pedestal **250** are fixed to each other via the double-sided adhesive tape **240** in the same manner as <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>. Then, the substrate of the other wire and the tensile device **260** are fixed to each other via the double-sided adhesive tape **240**. The dimensions of the double-sided adhesive tape **240** used in this case are the same as the dimensions described in <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>. In the case of FIG. **7**, the double-sided adhesive tape **240** is provided at a position at a certain distance from the end of the connecting member **222**, and the distance is the same as the distance described in <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>.

[0116] FIG. **8** is a schematic cross-sectional view of a fifth modification of the superconducting wire structure according to the first embodiment. In FIG. **8**, the connecting member **222** includes a third connecting member **222***c* and a fourth connecting member **222***d*.

[0117] The third connecting member **222***c* and the fourth connecting member **222***d* connect the joint layer **221** and at least one of the first wire **210** and the second wire **220**.

[0118] The fourth connecting member **222***d* is in contact with the other side surface of the connection layer **221** different from the side surface in contact with the third connecting member **222***c* in FIG. **8**. The fourth connecting member **222***d* covers the other side surface of the connection layer **221**, and thus, it is possible to further suppress the wear of the connection layer **221** due to the contact with the other superconducting wire structure **2008**, compared to FIG. **6**.

[0119] The superconducting wire structure **2008** in FIG. **8** can be manufactured as follows.

[0120] First, the first substrate **210***b* is disposed to be in contact with at least a portion of the third connecting member **222***c*. Thereafter, the joint layer **221** is applied on at least a part of the second superconducting layer **220***a*. Secondly, at least a part of the first superconducting layer **210***a*, and in the joint layer **221** at least a part of a surface on the opposite side to the surface with which the second superconducting layer **220***a* is in contact, are provided to face with each other. Thereafter, the second substrate **220***b* is disposed to be in contact with at least a portion of the fourth connecting member **222***d*. In this case, the connecting member **222***c* in contact with the first substrate **210***b* and the connecting member **222***d* in contact with the second substrate **220***b* can be formed using different materials or the same material.

[0121] Thus, a structure for obtaining the superconducting wire structure **2008** in FIG. **8** is obtained. Thereafter, the superconducting wire structure **2008** shown in FIG. **8** is obtained by the above-described pressurizing method and heat treatment.

[0122] The critical current, the mechanical strength, and the porosity of the joint layer **221** of the obtained superconducting wire structure **2008** of FIG. **8** can be measured in the same manner as the methods described in <Measurement of Critical Current>, <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and <Method for Measuring Porosity of Joint Layer>.

[0123] FIG. **9** is a schematic cross-sectional view of a sixth modification of the superconducting wire structure according to the first embodiment. In FIG. **9**, the superconducting wire structure **2009** further includes a third wire **230**.

[0124] The third wire **230** includes a third superconducting layer **230***a*, and a third substrate **230***b*. The third superconducting layer **230***a* is provided on the third substrate

230*b*. The third superconducting layer **230***a* and the third substrate **230***b* respectively may comprise either the same or different materials of the first superconducting layer **210***a* and the first substrate **210***b*.

[0125] The third superconducting layer **230***a* is provided to face the first superconducting layer **210***a* and the second superconducting layer **220***a*. By providing third wire **230**, the energy loss of the superconducting wire structure **2009** can be reduced. This is because when a current flows from the first wire **210** to the second wire **220**, the current can pass through the third superconducting layer **230***a* having a lower resistance.

[0126] The connecting member **222** is provided to connect the joint layer **221** to at least one of the first wire **210** and the second wire **220**. In FIG. **9**, the connecting member **222** is provided to cover the surface of the third wire **230** exposed to the outside of the third wire and the side surface of the joint layer **221**; however, for example, the connecting member **222** can be positioned between the first wire **210** and the second wire. In this case, the connecting member **222** is positioned between the first wire **210** and the second wire **220** and is provided to connect the joint layer **221** to at least one of the first wire **210** and the second wire **220**.

[0127] The connecting member **222** includes a region **223**. The region **223** is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the joint layer **221** in the first direction from the first wire **210** to the second wire **220**.

[0128] The superconducting wire structure **2009** in FIG. **9** can be manufactured as follows.

[0129] Firstly, the materials for the joint layer **221** are applied on the third superconducting layer **230***a* with a required thickness and are dried. After the materials are dried, they are baked in oxygen at equal to or more than 700° C. and equal to or less than 850° C., and the joint layer **221** is obtained. Secondly, each superconducting layer of the first wire **210** and the second wire **220** is arranged to face the same direction. Thereafter, the third wire **230** with the joint layer **221** is arranged to face the first superconducting layer **210***a* and the second superconducting layer **220***a* through the joint layer **221**. Lastly, the connecting member **222** is provided to be in contact with the third substrate **230***b*.

[0130] Thus, a structure for obtaining superconducting wire structure **2009** in FIG. **9** is obtained. Thereafter, the superconducting wire structure **2009** shown in FIG. **9** is obtained by the above-described pressurizing method and heat treatment.

[0131] The critical current, the mechanical strength, and the porosity of the joint layer **221** of the obtained superconducting wire structure **2009** of FIG. **9** can be measured in the same manner as the methods described above in <Measurement of Critical Current>, <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and <Method for Measuring Porosity of Joint Layer>.

[0132] A method for measuring the mechanical strength in FIG. **9** will be described with reference to FIG. **10**. Here, FIG. **10** is a schematic cross-sectional view showing a method of measuring the mechanical strength in the example of the superconducting wire structure **2009** shown in FIG. **9**.

<Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **9**>

[0133] In FIG. **10**, firstly, a case in which the third substrate **230***b* is covered with the connecting member **222** and the position of the third substrate **230***b* cannot be determined will be described. In this case, as shown in FIG. **10**, the third substrate **230***b* is fixed to the pedestal **250** via the double-sided adhesive tape **240** so that the entire connecting member **222** covering the third substrate **230***b* is fixed to the pedestal **250**. The first substrate **210***b* or the second substrate **220***b* is fixed to the tensile device **260** (in the example of FIG. **10** substrate **220***b* is fixed) in the same manner as the method described above in <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **6**>.

[0134] On the other hand, a case in which the third substrate **230***b* is not fully covered with the connecting member **222** and the position of the third substrate **230***b* can be determined in FIG. **10** will be described. In this case, the double-sided adhesive tape **240** is prepared in the same manner as <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and the third substrate **230***b* is fixed to the pedestal **250**. The first substrate **210***b* or the second substrate **220***b* is fixed to the tensile device **260** in the same manner as the method described in <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **6**>.

[0135] In the example of FIG. **10**, the tensile device **260** may be fixed to either one of the first and second substrate **210***b* or **220***b* or both of the first and second substrates **210***b* and **220***b*.

[0136] After the superconducting wire structure **2009** is fixed to the pedestal **250** and the tensile device **260** by using the double-sided adhesive tape **240**, the mechanical strength of the superconducting wire structure **2009** is measured by using the tensile device **260**.

[0137] The mechanical strength between the first wire **210**, the second wire **220**, or the third wire **230** and the joint layer **221**, can be evaluated by the above-described method.

[0138] FIG. **11** is a schematic cross-sectional view of a seventh modification of the superconducting wire structure according to the first embodiment.

[0139] The connecting member **222** includes a fifth connecting member **222***e* and a sixth connecting member **222***f*. In FIG. **11**, the fifth **222***e* is provided to cover the lower surface of the joint layer **221**.

[0140] The sixth connecting member **222***f* is provided to cover the surfaces of the third wire **230** exposed to the outside and the side surfaces of the joint layers **221**.

[0141] The third wire **230** is preferably covered with the connecting member **222**, which can improve the mechanical strength of the superconducting wire structure **2011**.

[0142] The superconducting wire structure **2011** in FIG. **11** can be manufactured as follows.

[0143] After a structure for obtaining the superconducting wire structure **2009** in FIG. **9** as described above is obtained, the connecting member **222***e* is arranged so that the connecting member **222***e* is in contact with at least a part of the first wire **210** and at least a part of the second wire **220**.

[0144] Thus, the structure for obtaining superconducting wire structure **2011** in FIG. **11** is obtained. Thereafter, the superconducting wire structure **2011** in FIG. **11** is obtained by the above-described pressurizing method and heat treatment.

[0145] The critical current, the mechanical strength, and the porosity of the joint layer **221** of the obtained superconducting wire structure **2011** of FIG. **11** can be measured in the same manner as the methods described above in <Measurement of Critical Current>, <Method for Measuring Mechanical Strength of Superconducting Wire Structure in FIG. **1**>, and <Method for Measuring Porosity of Joint Layer>.

[0146] A superconducting wire structure according to a first embodiment includes: a first superconducting wire; a second superconducting wire adjacent to the first superconducting wire; a joint layer electrically connecting the first superconducting wire and the second superconducting wire to each other; and a connecting member connecting the joint layer and at least one of the first superconducting wire and the second superconducting wire to each other. The connecting member has a melting point equal to or more than 900° C. and equal to or less than 1100° C. Thus, the superconducting wire structure having a high critical current and a high mechanical strength can be realized.

Second Embodiment

[0147] A superconducting coil according to a second embodiment includes the superconducting wire structure according to the first embodiment. Hereinafter, the description of a part of the content overlapping the first embodiment may be omitted.

[0148] FIG. **12** is a schematic perspective view of the superconducting coil according to the second embodiment. FIG. **13** is a schematic cross-sectional view of the superconducting coil according to the second embodiment.

[0149] A superconducting coil **700** according to the second embodiment is used, for example, as a magnetic field generating coil for a superconducting device, such as an NMR, an MRI, a heavy particle beam radiotherapy device, or a superconducting magnetic levitation railway vehicle.

[0150] The superconducting coil **700** includes a winding frame **110**, a first insulating plate **111***a*, a second insulating plate **111***b*, and a winding portion **112**. The winding portion **112** has a superconducting wire **120** and an inter-wire layer **130**.

[0151] FIG. **12** shows a state in which the first insulating plate **111***a* and the second insulating plate **111***b* are removed.

[0152] The winding frame **110** is formed of fiber-reinforced plastic, for example. The superconducting wire **120** has, for example, a tape shape. As shown in FIG. **12**, the superconducting wire **120** is wound around the winding frame **110** in a concentric so-called pancake shape with the winding central axis C as its axis.

[0153] The inter-wire layer **130** has a function of fixing the superconducting wire **120**. The inter-wire layer **130** has a function of suppressing destruction of the superconducting wire **120** due to vibration during use of the superconducting device or friction therebetween.

[0154] The first insulating plate **111***a* and the second insulating plate **111***b* are formed of fiber-reinforced plastic, for example. The first insulating plate **111***a* and the second insulating plate **111***b* have a function of insulating the winding portion **112** from the outside. The winding portion **112** is disposed between the first insulating plate **111***a* and the second insulating plate **111***b*.

[0155] The superconducting wire structure **200** according to the first embodiment is used as the superconducting wire **120**.

[0156] As described above, according to the second embodiment, a superconducting coil having improved characteristics can be realized by providing a superconducting wire having a low electrical resistance and a high mechanical strength.

Third Embodiment

[0157] A superconducting device according to a third embodiment is a superconducting device including the superconducting coil according to the second embodiment. Hereinafter, the description of a part of the content overlapping the first and the second embodiment will be omitted.

[0158] FIG. **14** is a block diagram of the superconducting device according to the third embodiment. The superconducting device according to the third embodiment is a heavy particle beam radiotherapy device **800**. The heavy particle beam radiotherapy device **800** is an example of the superconducting device.

[0159] The heavy particle beam radiotherapy device **800** includes a beam delivery system **50**, a synchrotron accelerator **52**, a beam transport system **54**, an emission system **56**, and a control system **58**.

[0160] The beam delivery system **50** has a function of, for example, generating carbon ions used for treatment and pre-accelerating the carbon ions to be incident on the synchrotron accelerator **52**. The beam delivery system **50** includes, for example, an ion source and a linear accelerator.

[0161] The synchrotron accelerator **52** has a function of accelerating the carbon ion beam from the beam delivery system **50** to an energy suitable for treatment. The superconducting coil **700** according to the second embodiment is used as the synchrotron accelerator **52**.

[0162] The beam transport system **54** has a function of transporting the carbon ion beam from the synchrotron accelerator **52** to the emission system **56**. The beam transport system **54** includes, for example, a deflection electromagnet.

[0163] The emission system **56** has a function of emitting the carbon ion beam from the beam transport system **54** to a patient that is an emission target. The emission system **56** includes, for example, a rotating gantry that allows the carbon ion beam to be emitted from any direction. The superconducting coil **700** according to the second embodiment is used as the rotating gantry.

[0164] The control system **58** controls the beam delivery system **50**, the synchrotron accelerator **52**, the beam transport system **54**, and the emission system **56**. The control system **58** is, for example, a computer.

[0165] In the heavy particle beam radiotherapy device **800** according to the third embodiment, the superconducting coil **700** according to the third embodiment is used as the synchrotron accelerator **52** and the rotating gantry. Therefore, the heavy particle beam radiotherapy device **800** with excellent characteristics is realized.

[0166] In the third embodiment, as an example of a superconducting device, the case of the heavy particle beam radiotherapy device **800** has been described. However, the superconducting device may be a nuclear magnetic resonance (NMR) apparatus, a magnetic resonance imaging (MRI) apparatus, or a superconducting magnetic levitation railway vehicle.

EXAMPLES

Example 1

[0167] Three wires on which a $GdBa_2Cu_2O_\delta$ layer (oxide superconducting layer) was formed were prepared. The width and length of the two wires were respectively 4 mm and 50 mm, and the width and length of the other wire were respectively 12 mm and 10 mm. 15 mm from the tip of the protective layers for the two long thin wires, and the entire surface of the protective layer for the short thick wire were wet etched using a mixed solution of nitric acid, ammonia, and hydrogen peroxide to expose the oxide superconducting layers (corresponding to the superconducting layers of the first and second wires and the superconducting layer of the third wire)

[0168] Next, powders of Gd_2O_3, $BaCO_3$, and CuO were prepared, appropriately weighed, and then sufficiently mixed. The mixed powder was subjected to a heat treatment at 900° C. to obtain a calcined body. The calcined body was pulverized, and the obtained powder was compression-molded to produce a green compact. The obtained green compact was sintered at 960° C., by which an oxide superconductor having a composition of $GdBa_2Cu_3O_\delta$ ($6\leq\delta\leq7$) was manufactured. The obtained oxide superconductor was wet pulverized and then a heat treatment in oxidizing atmosphere was performed at 470° C. to obtain a powder of monocrystalline or polycrystalline particles of $GdBa_2Cu_3O_\delta$ ($6\leq\delta\leq7$) having a mean particle size of 5.9 μm.

[0169] The obtained powder and a solution mainly including an organic compound containing the same metal element as that of the obtained powder were mixed at a weight ratio of 2:1, and the mixture was applied on the superconducting layer of the third wire. After drying the mixture at 150° C., the product was sintered in oxygen at 800° C.

[0170] Firstly, a 10 mm square Ag sheet having a thickness of 50 μm was placed on a pressurizing jig, and the first and the second wires were placed on the Ag sheet with the superconducting layers of each wire facing upward. Secondly, the third wire was placed on the superconducting layers of the first and the second wires with the joint layer facing downward, and another 10 mm square Ag sheet having a thickness of 50 μm was further placed on the third wire. Lastly, another pressurizing jig was placed on the Ag sheet in contact with the third wire. The superimposed wires and Ag sheets were interposed between pressurizing jigs from above and below and pressed with a torque of 4 N·m. A heat treatment was performed at 820° C. in argon containing 500 ppm oxygen, and subsequently another heat treatment was performed at 450° C. in oxygen while the wires and the Ag sheets were interposed between pressurizing jigs.

[0171] The sample taken out from the furnace was taken out from the pressurizing jig, and after terminals were attached, the sample was immersed in liquid nitrogen, and the critical current was measured. Another sample prepared in the same manner was used for the mechanical strength and then for the evaluation of the porosity.

<Method for Measuring Critical Current>

[0172] The critical current in the manufactured superconducting wire structure measured by a four terminal measurement method. The terminal was attached to the superconducting wire structure after the connection, and measurement was performed in liquid nitrogen. Since the voltage increases as the resistance increases, the current value at which the voltage starts to increase was defined as the critical current. The obtained values of critical current are shown in Tables 1 and 2 below. The values of critical current shown in Tables 1 and 2 are values indicating how many times they are as much as Comparative Example 1 when the value of Comparative Example 1 is defined as 1.

<Method for Measuring Mechanical Strength of Superconducting Wire Structure>

[0173] The mechanical strength of the superconducting wire structure was evaluated according to Japanese Industrial Standards JIS K 6849 (1994) "Testing methods for tensile strength of adhesive bonds (method of measuring by tensile load perpendicular to the bonding surface)". In the measurement, firstly, the third substrate was fixed to the pedestal via double-sided adhesive tape so that the entire member covering the third substrate was fixed to the pedestal.

[0174] Secondly, the first substrate was fixed to the tensile device via double-sided adhesive tape. The double-sided adhesive tape was prepared so that the width of the tape was 4 mm, which is the same length as the first wire. Further, the length of the tape was a length that enabled strength that did not cause peeling between the pedestal and the first substrate during measurement of the mechanical strength. A carbon tape for SEM observation was used for the double-sided adhesive tape.

[0175] Thus, as the operation of the tensile device, the tensile device was operated in a direction away from the first wire fixed to the tensile device via the double-sided adhesive tape. At this time, the operation speed of the tensile device was set to be constant, and the speed was set to be 700 μm/s. The distance by which the tensile device moved until the first wire was completely peeled off from the joint layer was measured as the mechanical strength. The obtained mechanical strengths are shown in Tables 1 and 2. The values of mechanical strength shown in Tables 1 and 2 are values indicating how many times they are as much as Comparative Example 1 when the value of Comparative Example 1 is defined as 1.

<Method for Measuring Porosity of Joint Layer>

[0176] First, the connection portion including the joint layer was embedded in an epoxy-based resin, and the vicinity of the center of the connection portion which was firmly connected was cut by a diamond saw because peeling was concerned at the end of the connection. At this time, the cross section for measuring the porosity of the joint layer was located at a position avoiding the connecting member existing between the joint layer and the first wire or the second wire.

[0177] Next, the cut surface was polished with waterproof paper and buffed, and then coated with a conductive material, and the observation positions were determined by SEM. Ion milling was performed on the determined observation positions, and SEM observation was performed after conductive coating was performed.

[0178] SEM images of three positions in the cross section were taken at a magnification of about 2500 times. At this time, the joint layer was processed to have three cross sections at positions where the sides corresponding to the long sides of the substantially quadrangular shape of the joint layer in plan view were equally divided into four. The SEM image was binarized to separate particles and voids using ImageJ, and then the areas of the particles and the voids were calculated. The porosity in each cross section was calculated from the ratio of the areas. The porosity obtained in each cross section was averaged to obtain the porosity of the joint layer to be obtained. The obtained porosities of the joint layer are shown in Tables 1 and 2.

<Method for Measuring Length Along First Direction of Member Entering Joint Layer>

[0179] The distance was measured by measuring the connecting member remaining on the first superconducting layer of the first wire after the separation of the joint layer and the first wire.

[0180] First, after <Method for Measuring Mechanical Strength of Superconducting Wire Structure> was performed, the joint layer and the first wire were separated from each other.

[0181] Thereafter, a mapping image of elements derived from the components of the connecting member was captured by EDX on the surfaces of the first superconducting layer of the first wire. This makes it possible to measure the depth of the connecting member entering the joint layer in the first direction.

[0182] When capturing EDX images, first, the area on the surface of the first superconducting layer, where the joint layer existed before separation from the first wire, was divided into five equal parts along the third direction, which intersects with the first direction and is parallel to the surface. The regions at both ends of the five equal parts along the third direction were excluded from the EDX imaging area. The EDX imaging magnification was set to a range where the connecting member with the longest distance along the first direction was within the field of view in each imaging area.

[0183] In each EDX image obtained in this way, the average length of the connecting member in the third direction on the surface of the first superconducting layer was determined. The average length of the invasion mentioned above was determined at positions dividing each width of the EDX images along the third direction into six equal parts. By averaging these five lengths obtained from each of the three EDX images, a total of 15 lengths, the length of the connecting member invading the joint layer along the first direction was determined. The obtained lengths along the first direction of the connecting member entering the joint layer are shown in Tables 1 and 2.

Example 2

[0184] A sample was manufactured and evaluated in the same manner as in Example 1, except that a powder of monocrystalline or polycrystalline particles of $GdBa_2Cu_3O_\delta$ ($6 \leq \delta \leq 7$) having a mean particle size of 2.2 μm was used, and the obtained powder and a solution mainly including an organic compound containing the same metal element as that of the obtained powder were mixed at a weight ratio of 1:1.

Example 3

[0185] A sample was manufactured and evaluated in the same manner as in Example 1, except that the thickness of the Ag sheet was 20 μm.

Example 4

[0186] A sample was manufactured and evaluated in the same manner as in Example 1, except that the Ag sheet was only placed under the first wire and the second wire.

Example 5

[0187] The obtained powder described in Example 1 and a solution mainly including an organic compound containing the same metal element as that of the obtained powder were mixed at a weight ratio of 2:1, and the mixture was applied on the Ag sheet instead of the third wire. After drying the mixture at 150° C., the product was sintered in oxygen at 800° C. A sample was manufactured and evaluated in the same manner as in Example 1, except that the obtained joint layer on the Ag sheet was placed on the first and the second superconducting layers and the other Ag sheet was not used.

Example 6

[0188] The first and the second wires with the protective layers separated described in Example 1 were only used as wires, the mixture described in Example 1 was applied on either of the wires. After drying the mixture at 150° C., the product was sintered in oxygen at 800° C. A sample was manufactured and evaluated in the same manner as in Example 1, except that the first wire and the second wire were arranged to face each other.

Example 7

[0189] A sample was manufactured and evaluated in the same manner as in Example 1, except that a Cu sheet was used instead of an Ag sheet.

Example 8

[0190] A sample was manufactured and evaluated in the same manner as in Example 1, except that an Au sheet was used instead of an Ag sheet.

Example 9

[0191] A sample was manufactured and evaluated in the same manner as in Example 1, except that a Ge sheet was used instead of an Ag sheet.

Example 10

[0192] A sample was manufactured and evaluated in the same manner as in Example 1, except that a powder of monocrystalline or polycrystalline particles of $GdBa_2Cu_3O_\delta$ ($6 \leq \delta \leq 7$) having a mean particle size of 1.7 μm was used, and the obtained powder and a solution mainly including an organic compound containing the same metal element as that of the obtained powder were mixed at a weight ratio of 1:1.

Example 11

[0193] A sample was manufactured and evaluated in the same manner as in Example 1, except that a powder of monocrystalline or polycrystalline particles of $GdBa_2Cu_3O_\delta$ ($6 \leq \delta \leq 7$) having a mean particle size of 8.6 μm was used, and the obtained powder and a solution mainly including an organic compound containing the same metal element as that of the obtained powder were mixed at a weight ratio of 1:2.

Example 12

[0194] A sample was manufactured and evaluated in the same manner as in Example 1, except that a powder of monocrystalline or polycrystalline particles of $GdBa_2Cu_3O_\delta$ ($6 \leq \delta \leq 7$) having a mean particle size of 1.3 μm was used, and the obtained powder and a solution mainly including an organic compound containing the same metal element as that of the obtained powder were mixed at a weight ratio of 1:1.

Example 13

[0195] A sample was manufactured and evaluated in the same manner as in Example 1, except that a powder of monocrystalline or polycrystalline particles of $GdBa_2Cu_3O_\delta$ ($6 \leq \delta \leq 7$) having a mean particle size of 11.0 μm was used, and the obtained powder and a solution mainly including an organic compound containing the same metal element as that of the obtained powder were mixed at a weight ratio of 1:2.

Comparative Example 1

[0196] A sample was manufactured and evaluated in the same manner as in Example 1, except that the Ag sheet was not placed.

Comparative Example 2

[0197] A sample was manufactured and evaluated in the same manner as in Example 2, except that the Ag sheet was not placed.

Comparative Example 3

[0198] A sample was manufactured and evaluated in the same manner as in Example 1, except that the Ag sheet was not placed and the entire connection portion was sealed with solder at 300° C. after the sample was sintered.

Comparative Example 4

[0199] A sample was manufactured and evaluated in the same manner as in Example 1, except that the Ag sheet was not placed and the entire connection portion was fixed by pressure bonding with Ag sheet at room temperature after the sample was sintered.

Comparative Example 5

[0200] A sample was manufactured and evaluated in the same manner as in Example 1, except that an Al sheet was used instead of the Ag sheet.

Comparative Example 6

[0201] A sample was manufactured and evaluated in the same manner as in Example 1, except that a Pt sheet was used instead of the Ag sheet.

[0202] The values of critical current, the mechanical strength, the porosity of the joint layer, and the length along the first direction of the connecting member entering the joint layer in all examples and comparative examples are shown in Tables 1 and 2.

TABLE 1

	Melting point of member [° C.]	Member	Porosity of joint layer [%]	Length of member entering joint layer [μm]	Critical current	Mechanical strength
Example 1	961.8	Ag	44	325	1.2	3.5
Example 2	961.8	Ag	36	277	1.9	3.6
Example 3	961.8	Ag	44	218	1.2	4.2
Example 4	961.8	Ag	44	230	1.2	3.3
Example 5	961.8	Ag	44	230	1.1	3.5
Example 6	961.8	Ag	44	325	1.2	3.5
Example 7	1085	Cu	44	38	1.2	3.3
Example 8	1064	Au	44	121	1.2	3.4
Example 9	938.2	Ge	44	782	1.2	3.7
Example 10	961.8	Ag	15	97	1.6	3.1
Example 11	961.8	Ag	65	421	1	4
Example 12	961.8	Ag	5	76	1.3	3
Example 13	961.8	Ag	80	596	1	4.1

TABLE 2

	Melting point of member [° C.]	Member	Porosity of joint layer [%]	Length of member entering joint layer [μm]	Critical current	Mechanical strength
Comparative Example 1	—	—	44	—	1	1
Comparative Example 2	—	—	36	—	1.2	1.1
Comparative Example 3	—	—	44	—	0.6	3.2
Comparative Example 4	—	—	50	—	0.1	3.5
Comparative Example 5	660	Al	44	1500	0.5	3.6

TABLE 2-continued

	Melting point of member [° C.]	Member	Porosity of joint layer [%]	Length of member entering joint layer [μm]	Critical current	Mechanical strength
Comparative Example 6	1769	Pt	44	0	1	1

[0203] In contrast to Comparative Examples 1 to 4 in which the connection started to peel off when the sample was taken out from the pressurizing jigs, all Examples 1 to 6 in which the Ag sheet was provided have improved characteristics of the critical current and the mechanical strength. In particular, in Example 2 in which a powder having a small mean particle size was used, the porosity of the joint layer was low, but the contact area between the particles or between the particle and the superconducting layer of the wire increased. Thus, the critical current was high in Example 2.

[0204] In Example 3 using the thin Ag sheet, the Ag easily entered the joint layer, and the mechanical strength was further improved.

[0205] On the other hand, in Comparative Example 3 in which the connection portion was sealed with solder, the powder in the joint layer and the superconducting layer in the wire were deteriorated by heat, and the mechanical strength was high, but the critical current was low. Similarly, in Comparative Example 4 in which the connection portion was fixed by pressure bonding with the Ag sheet at room temperature, the structure in the joint layer collapsed by the pressure, and the connection between the particles in the joint layer and the superconducting layer of the wire peeled off, and the mechanical strength was high, but the critical current was low.

[0206] It was found that the characteristics of the critical current and the mechanical strength were high also in Examples 7 to 9 in which Cu, Au, and Ge having a melting point of 900° C. to 1100° C. were used instead of the Ag sheet.

[0207] On the other hand, in Comparative Example 5 using the Al sheet having a melting point of less than 900° C., Al reacted with the GdBCO material in the joint layer, and the mechanical strength was high, but the critical current was low. In Comparative Example 6 using the Pt sheet having a melting point of more than 1100° C., it was found that the mechanical strength was low because the Pt sheet was not softened.

[0208] In the Examples 10 and 11 in which the porosities were 10% to 70%, the characteristics of the critical current and the mechanical strength were high, and the characteristics in Example 12 where the porosity was less than 10% and in Example 13 where the porosity was more than 70% were high as well.

[0209] In addition, in Comparative Example 5 using an Al sheet having a melting point of 660° C. instead of the Ag sheet, it was found that the melting of the connecting member excessively progressed, and the length of the connecting member entering the joint layer was increased compared to Example 1. Thus, the connecting member pushed the particles composing the joint layer and this made it unable to obtain a sufficient connection area and to provide a sufficient current path. Hence, it was found that the critical current in Comparative Example 5 was lower than Example 1.

[0210] In addition, in Comparative Example 6 in which the Pt sheet having a melting point of 1769° C. was used instead of the Ag sheet, the connecting member was not sufficiently melted, and the connecting member did not enter the joint layer. It was found that the mechanical strength as the superconducting wire structure in Comparative Example 6 was lower than Example 1.

[0211] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the superconducting layer joint structure, the superconducting wire structure, the superconducting coil, and the superconducting device described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the devices and methods described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

[0212] Hereinafter, some aspects according to the embodiments will be described.

[0213] [1] A superconducting wire structure, comprising:

[0214] a first superconducting wire;

[0215] a second superconducting wire adjacent to the first superconducting wire;

[0216] a joint layer electrically connecting the first superconducting wire and the second superconducting wire to; and

[0217] a connecting member connecting the joint layer and at least one of the first superconducting wire and the second superconducting wire,

[0218] wherein the connecting member has a melting point equal to or more than 900° C. and equal to or less than 1100° C.

[0219] [2] The superconducting wire structure according to [1],

[0220] wherein a porosity of the joint layer is equal to or more than 10% and equal to or less than 70%.

[0221] [3] The superconducting wire structure according to [1] or [2],

[0222] wherein the connecting member includes at least one selected from: a first region existing between the joint layer and the first superconducting wire; and a second region existing between the joint layer and the second superconducting wire,

[0223] the first region and the second region satisfy at least one of a first condition and a second condition,

[0224] in the first condition, in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the first region is located in a range of equal or more than 5 um and equal or less than 1 mm from an end of the first superconducting wire,

[0225] in the second condition, in the first direction, the second region is located in a range of equal or more than 5 um and equal or less than 1 mm from an end of the second superconducting wire.

[0226] [4] The superconducting wire structure according to [1] or [2],

[0227] wherein in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the connecting member is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the joint layer.

[0228] [5] The superconducting wire structure according to any one of [1] to [4],

[0229] wherein the connecting member includes at least one selected from the group including: Ag, Cu, Au, and Ge.

[0230] [6] The superconducting wire structure according to any one of [1] to [5],

[0231] further comprising a third superconducting wire,

[0232] wherein the joint layer is located between the first superconducting wire and the third superconducting wire, and between the second superconducting wire and the third superconducting wire,

[0233] the joint layer electrically connects the first superconducting wire and the third superconducting wire, and the second superconducting wire and the third superconducting wire.

[0234] [7] The superconducting wire structure according to [6],

[0235] wherein a porosity of the joint layer is equal to or more than 10% and equal to or less than 70%.

[0236] [8] The superconducting wire structure according to claim [**6**] or [**7**],

[0237] wherein the connecting member includes at least one selected from: a first region existing between the joint layer and the first superconducting wire; and a second region existing between the joint layer and the second superconducting wire,

[0238] the first region and the second region satisfy at least one of a third condition and a fourth condition,

[0239] in the third condition, in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the first region is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the first superconducting wire,

[0240] in the fourth condition, in the first direction, the second region is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the second superconducting wire.

[0241] [9] The superconducting wire structure according to any one of [6] to [8],

[0242] wherein in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the connecting member is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the joint layer.

[0243] [10] The superconducting wire structure according to any one of [6] to [9],

[0244] wherein the connecting member includes at least one selected from the group including: Ag, Cu, Au, and Ge.

[0245] [11] A superconducting coil, comprising:

[0246] the superconducting wire structure according to any one of [1] to [10].

[0247] [12] A superconducting device, comprising:

[0248] the superconducting coil according to [11].

What is claimed is:

1. A superconducting wire structure comprising:

a first superconducting wire;

a second superconducting wire adjacent to the first superconducting wire;

a joint layer electrically connecting the first superconducting wire and the second superconducting wire; and

a connecting member connecting the joint layer and at least one of the first superconducting wire and the second superconducting wire,

wherein the connecting member has a melting point equal to or more than 900° C. and equal to or less than 1100° C.

2. The superconducting wire structure according to claim **1**,

wherein a porosity of the joint layer is equal to or more than 10% and equal to or less than 70%.

3. The superconducting wire structure according to claim **1**,

wherein the connecting member includes at least one selected from: a first region existing between the joint layer and the first superconducting wire; and a second region existing between the joint layer and the second superconducting wire,

the first region and the second region satisfy at least one of a first condition and a second condition,

in the first condition, in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the first region is located in a range of equal or more than 5 um and equal or less than 1 mm from an end of the first superconducting wire,

in the second condition, in the first direction, the second region is located in a range of equal or more than 5 um and equal or less than 1 mm from an end of the second wire.

4. The superconducting wire structure according to claim **1**,

wherein in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the connecting member is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the joint layer.

5. The superconducting wire structure according to claim **1**,

wherein the connecting member includes at least one selected from the group including: Ag, Cu, Au, and Ge.

6. The superconducting wire structure according to claim **1**,

further comprising a third superconducting wire,

wherein the joint layer is located between the first superconducting wire and the third superconducting wire, and between the second superconducting wire and the third superconducting wire, the joint layer electrically connects the first superconducting wire and the third superconducting wire, and the second superconducting wire and the third superconducting wire.

7. The superconducting wire structure according to claim **6**,

wherein a porosity of the joint layer is equal to or more than 10% and equal to or less than 70%.

8. The superconducting wire structure according to claim **6**,

wherein the connecting member includes at least one selected from: a first region existing between the joint layer and the first superconducting wire; and a second region existing between the joint layer and the second superconducting wire,

the first region and the second region satisfy at least one of a third condition and a fourth condition,

in the third condition, in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the first region is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the first wire,

in the fourth condition, in the first direction, the second region is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the second wire.

9. The superconducting wire structure according to claim **6**,

wherein in a first direction crossing a direction from the first superconducting wire to the joint layer and along the first superconducting wire, the connecting member is located in a range of equal or more than 5 um and equal or less than 1 mm from an edge of the joint layer.

10. The superconducting wire structure according to claim **6**,

wherein the connecting member includes at least one selected from the group including: Ag, Cu, Au, and Ge.

11. A superconducting coil, comprising:

the superconducting wire structure according to claim **1**.

12. A superconducting device, comprising:

the superconducting coil according to claim **11**.

* * * * *